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Paleomagnetism of the Zambales ophiolite, Luzon, northern Philippines

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Abstract

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Paleomagnetic results are reported from 52 sites in the Zambales ophiolite. The sites provide vertical sampling from the overlying sediments, in the north and south, to layered gabbros and dunites in the Acoje and Coto mines, in the west. They extend over an area of approximately 100 by 50 km.

A Late Miocene–Pliocene intrusion in the Tarlac region gives a direction marginally distinguishable from the present field indicating a small clockwise rotation. This is consistent with other indications of clockwise rotation since Late Miocene in Luzon. Early and Mid-Miocene sediments from Barlo in the north and Tarlac in the south indicate a period of rapid counterclockwise rotation during the Miocene, again consistent with other results from Luzon. Eocene–Oligocene sediments overlying the ophiolite in the north and south indicate further counterclockwise rotation with northerly translation. The interpretation of the directions in the ophiolite itself is more problematical. Directions similar to those in the Eocene–Oligocene and in Mid-Miocene sediments probably reflect secondary magnetizations and magnetization associated with the unroofing of the ophiolite. We interpret the most strongly counterclockwise rotated magnetizations represented by a 44 Ma sill in the Barlo area and of the 46 Ma dykes at Coto as the primary direction of the ophiolite.

The counterclockwise rotation and northward translation interpreted from our results is consistent with the Holloway model of the origin of the Philippines. However, this model neglects the role of the Philippine Sea Plate in the tectonic history of the Philippines. The counterclockwise rotation is also consistent with rotation of Luzon in the left-lateral shear between the Philippine Sea Plate and Asia, as suggested by Sarewitz and Karig. Our data do not distinguish between counterclockwise rotation of a single block, or of a number of smaller blocks. The data suggest post-Late Miocene clockwise rotation for Luzon, when it was coupled to the Philippine Sea Plate. Our data are not consistent with a Philippine Sea Plate origin for Zambales.

Introduction

The Philippine archipelago is a geologically complex assemblage of ophiolites, arc related materials and at least one continental fragment (e.g., Karig, 1983). From whence these various terranes came and how they were assembled into their present configuration is not understood. Paleomagnetism has proved a valuable tool in answering such questions and this study is aimed at providing tests of the various models of the geologic evolution of the Philippines.

The area investigated is in northwestern Luzon and is dominated by the Zambales ophiolite. By documenting its latitude of origin and subsequent history of movement, the paleomagnetism of the ophiolite and of its overlying carapace of sediments could play an important role in unravelling

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the tectonics of the region. The Zambales ophiolite is a particularly attractive paleomagnetic target because it is extensive and yet comparatively intact.

Paleomagnetism has proved difficult in the Philippines. First, the location of successive arcs in Luzon gives rise to repeated intrusions with associated hydrothermal events, causing widespread secondary magnetizations. Second, the tectonic complexity of the region coupled with limited outcrop has led to uncertainty in tectonic corrections, the absence of fold tests and a general lack of structural control. Third, the low inclination at equatorial latitudes makes the demonstration of the sense of rotations more difficult than at higher latitudes. A more detailed discussion of the interpretation of the paleomagnetic data is therefore given than is customary.

Geology of the Zambales ophiolite

In a volume devoted to the Zambales ophiolite there is little need for a lengthy discussion of its geology in a paper devoted to its paleomagnetism. The reader can go elsewhere in this volume for recent views. Earlier descriptions were given by Corby (1951) and Rossman (1964). Figure 1, which is from Hawkins and Evans (1983), illustrates the overall geology of the ophiolite.

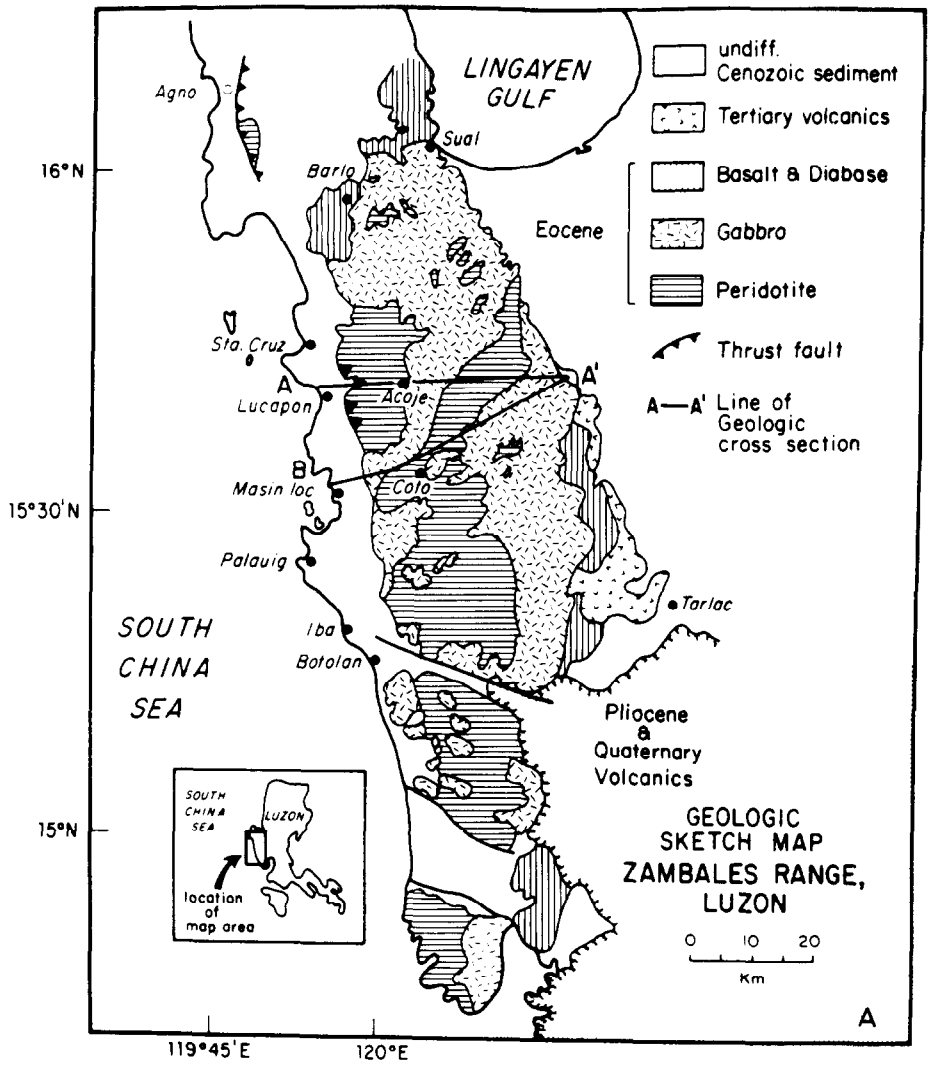
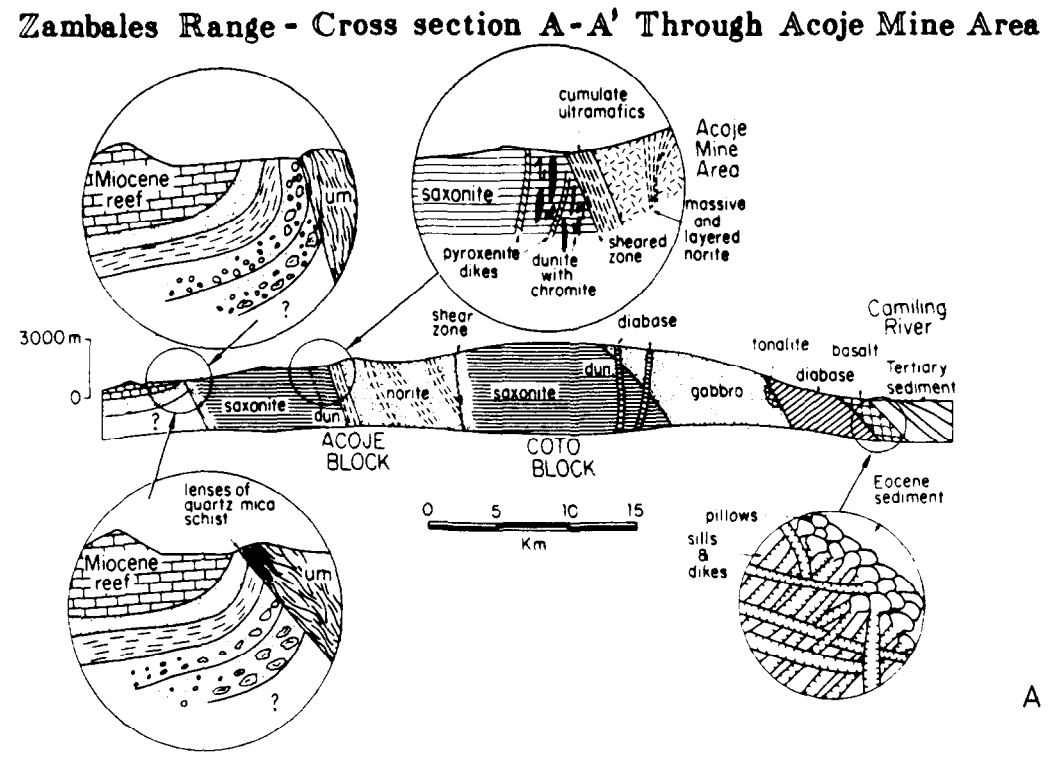


Fig. 1. Geological map and cross sections of Zambales ophiolite (Hawkins and Evans, 1983).



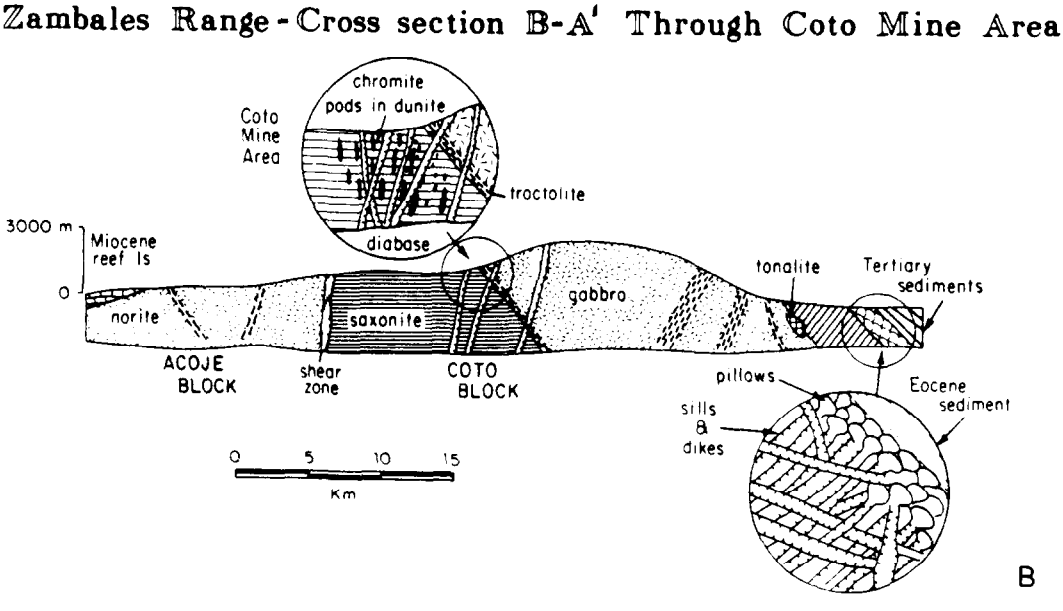


Fig. 1 (continued).

The age of the ophiolite has been established both from the age of the overlying sediments and by radiometric age determinations. Ages obtained from the ophiolite are given by Wolfe (1981) in his review of geochronology in the Philippines. We have obtained three additional, whole rock K/Ar,

ages of units within the ophiolite which are critical to the interpretation of the paleomagnetism. The first was for a granodioritic phase alongside the road from Alaminos to Sual at 16°05′N, 120°02′E. This yielded 44 ± 3.5 Ma. The second was for a small sill among the pillow lavas near Barlo Minc at 15°59′N, 119°58′E, which gave 44.1 ± 3.0 Ma. The third was from a dyke which intrudes the gabbros in the Coto Mine area at 15°35′N, 120°06′E and has an age of 46.6 ± 5.1 Ma. These ages are consistent with other radiometric data and indicate an Eocene age for the ophiolite, but more refined age determinations are required.

Hawkins and Evans (1983) have described the geochemistry of the ophiolite in detail and have established that the body has both arc- and ocean-floor affinities. The Coto block in the south is typical of back arc and Acoje in the north resembles island arc (Fig. 1). Elsewhere in this volume, Geary et al. (1989) have proposed a two stage crustal history for the Zambales ophiolites with an initial N-MORB type ocean crust, later modified by incipient island arc magmatism. Thus, while Hawkins and Evans (1983) see separate units which come from different locations, Geary et al. (this vol.) suggest that within a single region back-arc ocean floor was the site of later arc activity. Geary and Kay (1983) studied the petrology of the igneous units and interpreted their metamorphism in terms of ocean-floor hydrothermal activity with the temperature gradient beneath the sea floor modulating its severity.

The overlying sediments are best seen in the east. These sediments have been studied by Garrison et al. (1979), Schweller et al. (1983) and Bachman et al. (1983). Hemipelagic Eocene sediments of the lowermost Aksitero Formation can be traced upward into sediments with increasing volcanic input and eventually into the coarse sandstones and conglomerates of the Moriones Formation. A similar succession is seen in the north. In the west, the base of the ophiolite is faulted against Miocene sediments containing clasts from the ophiolite. Clearly the ophiolite had been emplaced, uplifted, and was being eroded by this time. The overall eastward dip of the body is reflected by the progression from tectonized peridotite in the west to the sills and flows in the east. However, the uppermost units are also seen in the northwest. The fieldwork of Rossman (1964) suggests that we may be seeing the eastern half of a dome like structure.

The regional tectonic setting of the Zambales ophiolite has been touched upon by many authors in more general discussions of the Philippines, the Philippine Sea Plate and the South China Sea (e.g., Gervasio, 1971; Hamilton, 1979; Holloway, 1982; McCabe et al., 1982; Karig, 1983; Sarewitz and Karig, 1986; Mitchell et al., 1986).

A key step in understanding the region was taken by Taylor and Hayes (1983), who documented the timing of the opening of the South China Sea. Recognition of the sea-floor anomalies demonstrated that the spreading took place between 32 and 18 Ma. This demonstrates that the Zambales ophiolite cannot be from the South China Sea. It is at least 10 Ma older than the oldest South China Sea floor. Recently, reinvestigation of the South China Sea by Pautot et al. (1986) suggests that the orientation of spreading for the final spreading phase was NW–SE. This is more consistent with the extrusion tectonics of Molnar and Tapponnier (1975), than the N–S spreading.

Recent models of the tectonic history of the Philippines can be divided into three types. One follows the ideas of Holloway (1982). Briefly, the Philippine archipelago archipelago is considered to be initiated as an Early Tertiary E–W striking arc at equatorial latitudes. This arc rotated counterclockwise (Fig. 2) and eventually collided with the Palawan microcontinent, as it was carried south by South China Sea spreading. This collision with the Palawan microcontinent may have stopped the South China Sea spreading. Such a model requires counterclockwise rotations and northerly displacement for the Philippines during the Late Eocene and Oligocene. In addition to this, during the Miocene collision Zambales would have been further rotated. As was noted by Holloway, this is consistent with the early results reported by Hsu (1972) and by Dunn et al. (1978). The model ignores interaction with the Philippine Sea Plate, which was also moving northward at this time.

Another view of Philippines tectonic history

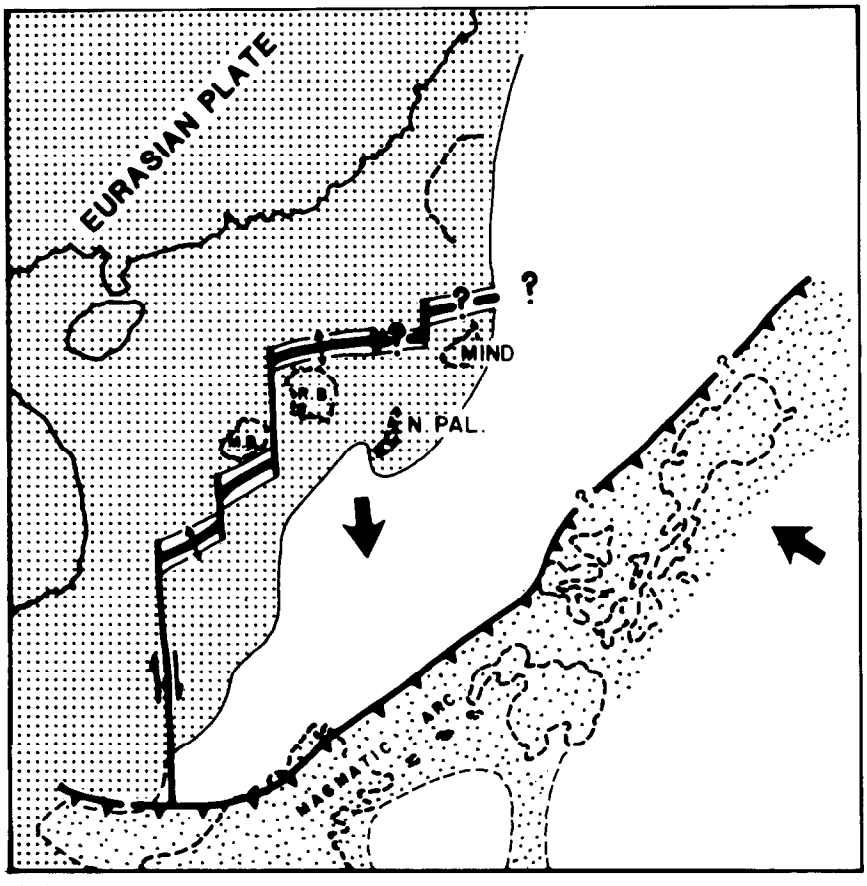


Fig. 2. Tectonic evolution of the Philippines, Eocene paleogeographic reconstruction (Holloway, 1982). A proto-Philippine arc continuing westward through Palawan to Borneo is shown rotating counterclockwise. The model postulates a collision with rifted fragments from China and the arc. It does not consider the interaction with the Philippine Sea Plate which is also moving northwards, but rotating clockwise.

has been proposed by Karig (1983) and by Sarewitz and Karig (1986). An important feature of their studies has been the recognition of large-scale strike-slip deformation within the Philippines (Fig. 3). They suggest that the Philippines have had a long association with the Philippine Sea Plate, which is in contrast to Holloway (1982). The strike slip motion is an expression of the deformation within a major sinistral shear zone between the clockwise rotating Philippine Sea Plate and the margin of Asia.

Still another view has been proposed by McCabe et al. (1987), who place the Philippines on the advancing front of the Philippine Sea Plate, thereby accounting for the interpreted east dipping subduction in the western Philippines. Eventually convergence between the Philippine Sea Plate and the Palawan microcontinent brings about the same collision as proposed by Holloway (1982). Indeed McCabe et al. (1982) interpreted paleomagnetic data as evidence of this collision.

The different models of the tectonic history of the Philippines make predictions which should be distinguishable paleomagnetically, although elements of all three models may have validity. It is the long term aim of our work to provide critical paleomagnetic data and so test the models of Philippine geology.

Previous paleomagnetic studies

The pioneering paleomagnetic studies of the Philippines were carried out by Hsu (1972). The principal results were that Luzon and other parts of the northern Philippines had moved northward and rotated counterclockwise during the Tertiary.

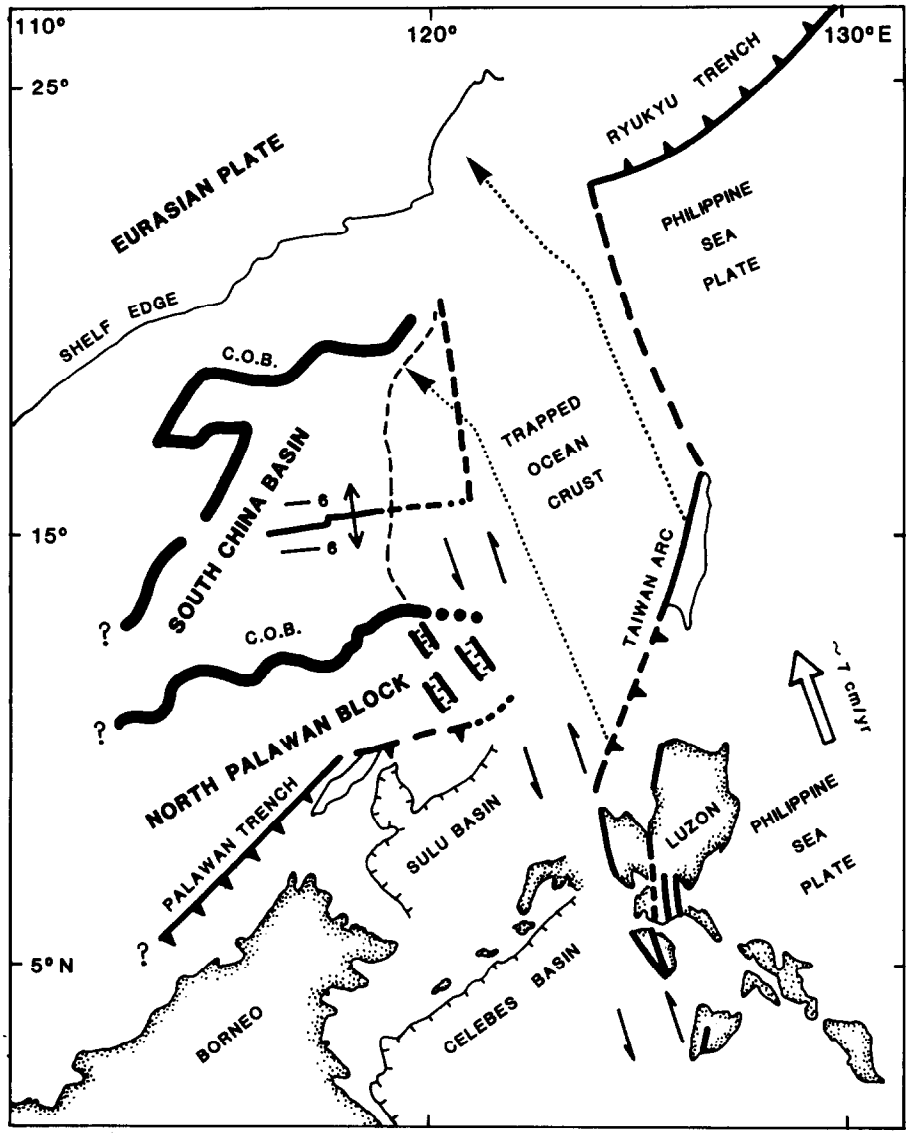


Fig. 3. Tectonic evolution of the Philippines, Early Miocene paleogeographic reconstruction (Sarewitz and Karig, 1986). In this model the Philippine archipelago is decoupled from Palawan and Borneo. NNW–SSE left-lateral shears between the Philippine Sea Plate and Asia accommodate relative motion and the Philippines are deformed within this shear zone.

A more recent study in the Bataan region by De Boer et al. (1980) suggested that Luzon had rotated by as much as 35° counterclockwise in the past 4 M.y. This is not consistent with the earlier results of Hsu (1972).

McCabe et al. (1982) made a paleomagnetic study in the island of Panay and reported data from six Cenozoic sites, which showed easterly declination anomalies of about 20°. Comparing these with the data to the north in Marinduque and Luzon, which had westerly declination anomalies, they argued that the pattern of declination anomalies was due to the collision of the Palawan microcontinent with the Philippines. In this they followed Holloway (1982) and provided paleomagnetic confirmation for the model.

Results from 50 sites in Cenozoic rocks from the Northern Philippines were reported by Fuller

et al. (1983). The principal results were as follows:

(1) No rotations or translations were detected in Plio-Pleistocene data.

(2) A possible clockwise rotation of Late Miocene sites was seen the Baguio region.

(3) A counterclockwise rotation of Early–Mid-Miocene sites with no major N–S translation was interpreted in both the Baguio and Zambales region.

(4) Eocene and Oligocene sites from the Zambales ophiolite and overlying sediments showed major northward motion and were interpreted to be rotated counterclockwise.

A simple model was constructed which called upon the buttressing effect of the Palawan microcontinent to account for the counterclockwise rotation of Luzon. This again was consistent with the Holloway model.

Additional results were presented in Fuller (1985) and an interpretation of the Late Miocene rotation given in terms of attachment to the Philippine Sea Plate. A transform boundary, or a region of diffuse deformation, was postulated immediately to the south of Luzon to accommodate the differential rotation between Luzon and the southern and central Philippines. Such a model assumes that the southern and central Philippines were held between the west-dipping Philippine subduction zone and the east-dipping subduction zones of Cotabato and Mindanao. However, prior to the formation of the East Luzon trench, Luzon was a part of the Philippine Sea Plate. As the Luzon trench propagated to the north, some form of adjustment must have taken place to the west as the plate boundary between Philippine Sea Plate and the South China Sea moves north. It is not clear how exactly this deformation is taken up. V. Courtillot (pers. commun., 1987) has suggested that there may be some relevant analogies between overlapping spreading centers and overlapping subduction zones. The shallow diffuse seismicity suggests that the plates are deforming internally (D.R. Sarewitz, pers. commun., 1988).

The most recent paleomagnetic study of the Philippines to be reported is McCabe et al. (1987). It is principally concerned with the central Philippines to the south of Luzon. However, some results are presented from Luzon and it is asserted that the sense of the Luzon rotation is clockwise. This interpretation of the Luzon data is inconsistent with the earlier interpretations of Hsu (1972) and Fuller et al. (1983) and indeed with that of McCabe et al. (1982). In this new view of the Luzon data proposed by McCabe et al. (1987), they no longer provide evidence of the Holloway collision model. Clearly this is a major revision of the interpretation of the paleomagnetism of Luzon. It is contested below.

Paleomagnetic methods

For the most part standard paleomagnetic methods have been used. Six to ten samples were drilled per site and oriented by magnetic, or sun, compass. The magnetic measurements were made in a shielded room, so the samples were never exposed to a field much in excess of 200 nanoteslas (gammas) throughout analysis. Remanent magnetization was measured either with a Molspin spinner magnetometer, or a 2G RF driven SQUID magnetometer with horizontal access.

Both thermal and AF demagnetization were used. The latter was carried out with a Schonstedt demagnetizer with demagnetization along each of the Cartesian coordinates at each step. The sequence of directions used was varied in a nonsystematic way. Thermal demagnetization was carried out both by the standard stepwise method and by a method in which measurements are made at high temperature developed in our laboratory (Dunn and Fuller, 1984). On line plots are used with both methods.

To isolate different components of magnetization, line fitting was used as well as difference vector analysis. At a small number of sites great circle analysis (Halls, 1976) revealed directions of magnetization which could not be obtained as stable end points of demagnetization. The statistical methods of Kirschvink (1980) were used. Site means were calculated by the method of Fisher (1953). A feature of paleomagnetism in the Philippines is the large amount of secondary magnetization encountered. In view of this we have cited more than one direction for sites, at which more than stable component was isolated.

In the face of the pervasive secondary magnetizations in these rocks, an attempt has been made to use AF demagnetization characteristics to identify the origin of the NRM. It is clear from standard analysis that the magnetization is carried by magnetite, the problems are to establish when it was acquired and what its significance is. The particular method utilizes plots of the ratio of NRM to saturation isothermal remanence (IRM (s)) during AF demagnetization (Fuller et al., 1988). The aim of the method is to provide an indication of the validity of the NRM from the characteristics of the magnetization, rather than simply by relying upon agreement with other results prejudged to be correct.

Paleomagnetic results

The principal paleomagnetic collections have been made from sites near Sual and at the Barlo Mine in the north, at the Coto and Acoje mines on the west side of the ophiolite, and along the Moriones, Camelling and Aksitero rivers in the southeast. The collections from the north are from the uppermost gabbros, the sills and flows as well as sediments overlying the ophiolite. At the Acoje and Coto mines the deepest part of the ophiolite was sampled. In the southeast, the various river sections provide good outcrops of the sediments on top of the ophiolite and of the flows, sills, and uppermost gabbros. In addition there are some younger intrusives of probable Late Miocene–Pliocene age (E.E. Geary, pers. commun., 1987).

The results are described by localities, starting in the north and proceeding counterclockwise around the body ending in the southeast outcrops near Tarlac. Smaller preliminary collections have been made at Olongapo in the south and near Botolan on the west side. However, the data are

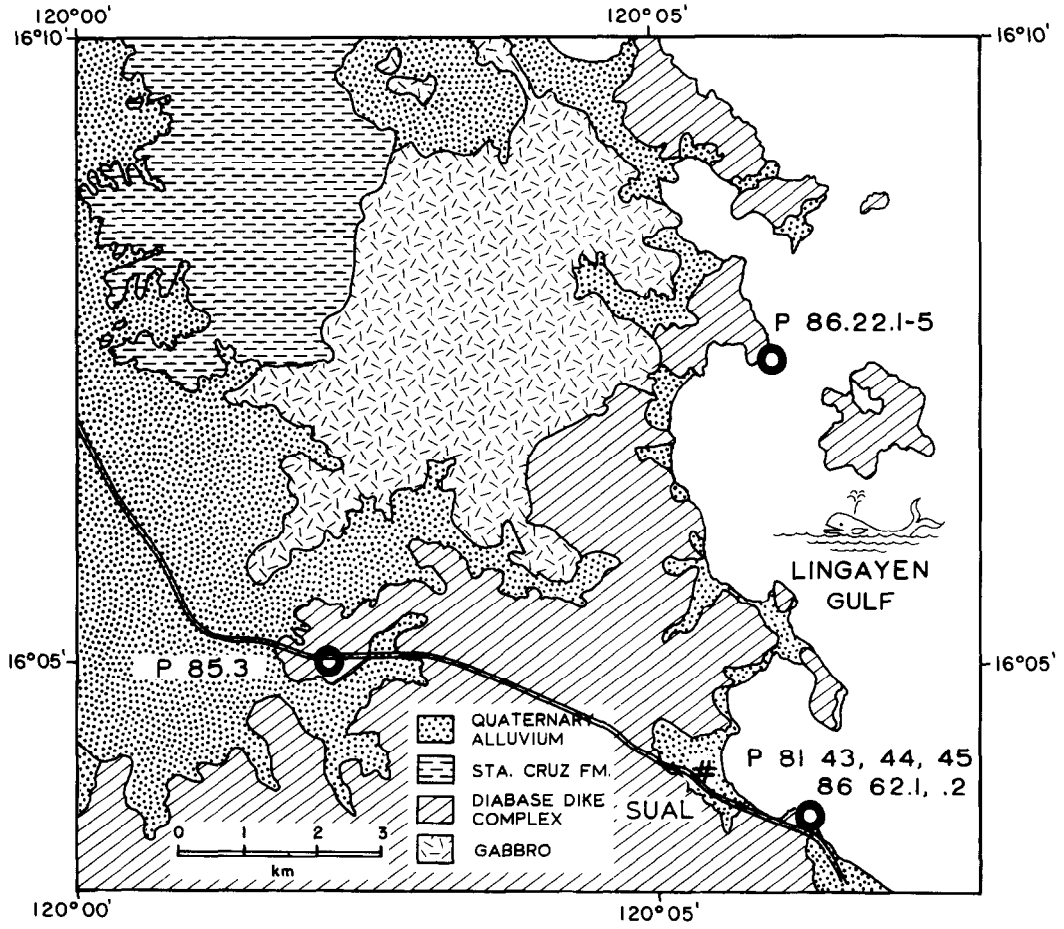


Fig. 4. Map of Sual area sites (geology from Philippines Bureau of Mines and Geo-sciences—Lingayen quadrangle map).

insufficient to warrant detailed discussion. In view of the obvious difficulties in assigning tectonic corrections deep in ophiolite bodies, all results are presented initially without a tectonic correction. The tectonic corrections are then discussed site by site.

Sual sites

In the extreme north the uppermost units of the ophiolite are found in the Lingayan quadrangle (Fig. 4). Samples were obtained from flows and dykes at the most northerly site P86.22. To the south, in Sual itself, the uppermost gabbros and sills are exposed at another beach site. Along the road to Alaminos, to the west of Sual, a granodioritic phase is seen in the gabbros with associated dykes. This granodiorite yielded a K/Ar, whole rock, age of 44 ± 3.5 Ma. A tectonic correction for these sites comes from the top surfaces of flows at the northernmost site.

Results

The mean direction of declination 151.2° and inclination 3.8° for the first site (P86.22.1) was obtained using the method of convergent great circles (Halls, 1978; Kirschvink, 1980). The various demagnetization paths were sufficiently dissimilar that the method worked satisfactorily on a single flow.

There are samples from this site with a single component magnetization differing from the direction obtained from the Halls analysis. Zijderveld plots reveal a direction with a declination of 112° and inclination of +19° which is unaffected by thermal demagnetization from about 350 to 550°C. This direction is recognized as a second direction found at the site.

Sites P86.22.2–4, at this locality, are in individual small dykes. In general, thermal demagnetization proved more effective in separating stable components of NRM than did AF demagnetization. The dyke P86.22.2 gave consistent directions with good convergence on the Zijderveld plots. Similarly a second dyke P86.22.3 also gave consistent directions throughout much of the blocking temperature range. A third dyke P86.22.4 did not give such consistent stable end points, but a mean declination and inclination of 96° and +14° was obtained from the directions in the highest AF and thermal demagnetization range.

A second flow (P86.22.5) gave stable endpoints on thermal and AF demagnetization, which defined a range in declination from 82° to 130°. In contrast, the inclination was relatively consistent. The data from this flow were not suitable for Fisher analysis and have not been included in our final analysis.

The results from site P86.22 define a considerable range in declination, but the inclination is

TABLE 1

Summary of results from Sual

Site No.	Unit	Location Lat. (°)(′)	Location Long. (°) (′)	N	Demag. AF/TH	Inc. (°)	Dec. (°)	R	k	α_{95}	Struct. corr.	Inc′. (°)	Dec′. (°)
Eocene–Oligocene													
86.22.1	flow	16 08	120 06	7	AF/TH	+3.8	151.2	–	–	6.6*	300/40N	+22.5	142.5
86.22.2	dyke	16 08	120 06	4	T400–550	+3.6	141.9	4.00	1040.1	2.9	300/40N	+16.7	134.8
86.22.3	dyke	16 08	120 06	4	T300–560	−1.1	102.6	3.96	81.8	10.2	300/40N	−11.9	107.2
86.22.4	dyke	16 08	120 06	5	T400–550	+14.0	096.3	4.95	74.5	8.9	300/40N	−3.7	092.9
81.43	sill	16 04	120 06	5	A200	+8.2	135.7	4.88	32.4	13.6	300/40N	+16.3	126.8
81.44	sill	16 04	120 06	4	A200	−11.7	132.2	3.93	41.5	14.4	300/40N	−1.3	136.8
86.62.2	sill	16 04	120 06	4	T400–550	−18.9	128.9	3.94	53.8	12.6	300/40N	−8.9	138.9
81.45	gabbro	16 04	120 06	5	A200	−11.8	127.0	4.87	31.4	13.9	300/40N	−4.6	132.9
86.62.1	gabbro	16 04	120 06	6	T500–550	−17.4	133.0	5.94	81.6	7.5	300/40N	−8.2	139.0
85.3	gdior.	16 05	120 02	11	A200–500	+3.8	130.3	10.90	103.6	4.5	300/40N	+9.5	125.5

consistently relatively shallow. A puzzling feature is that the range in declination seen within individual sub-sites is comparable with the between subsites range. Yet, if the samples from each of the dykes of flows are carrying a primary TRM, they should be recording a single spot reading of the field.

The second site sampled is on the beach at Sual. This site was sampled in 1979 and again in 1986. Consistent results were obtained on each occasion from the gabbros. The behavior of the three sites in the sills is less straightforward, but stable endpoints could be obtained from several samples from each site. The results are given in Table 1.

The final site in the Sual region was in a small granodioritic phase in the sheeted dyke complex. Both the intrusive and dykes gave consistent stable end points for both thermal AF demagnetization (Table 1).

Figure 5 summarizes the results from Sual. They are all of a single polarity with the considerable spread in declination and a consistently shallow inclination. If the tectonic correction of 300°–40° NE, obtained from the top surface of the lavas is used, the results are very little changed and the final mean has a declination of 132° and an inclination of +2°.

NRM versus IRM (s) demagnetization analysis was carried out for various samples from the Sual sites. The initial ratios of NRM : IRM (s) are invariably less than 10^{-2}. Numerous samples from the flows, dykes and the gabbros have concave upward slopes indicating that in the lowest ranges of AF demagnetization the NRM is being demagnetized more rapidly than the IRM (s). This is not characteristic of a thermoremanent magnetization carried by fine grain magnetite. There is a wide range in both NRM: IRM (s) values and also in similar ratios after 200 Oe AF demagnetization, indicating again that the magnetization cannot be a simple primary TRM.

Interpretation

There is a number of aspects of the magnetization of these Sual sites which suggests that secondary magnetizations may be playing an important role. A single polarity is seen from a large number of sites, whose magnetization was unlikely to be acquired within a single reversal epoch. The magnetization is not dominated by fine stable carriers. The spread in declination is large and within site dispersion is often comparable to that between sites. However, thin-section examination gave no evidence of alteration apart from that which probably took place on the sea floor (Geary and Kay, 1983).

The two extremes of the declination range are approximately 130° and 80°. From these sites alone, we cannot distinguish which, if either of

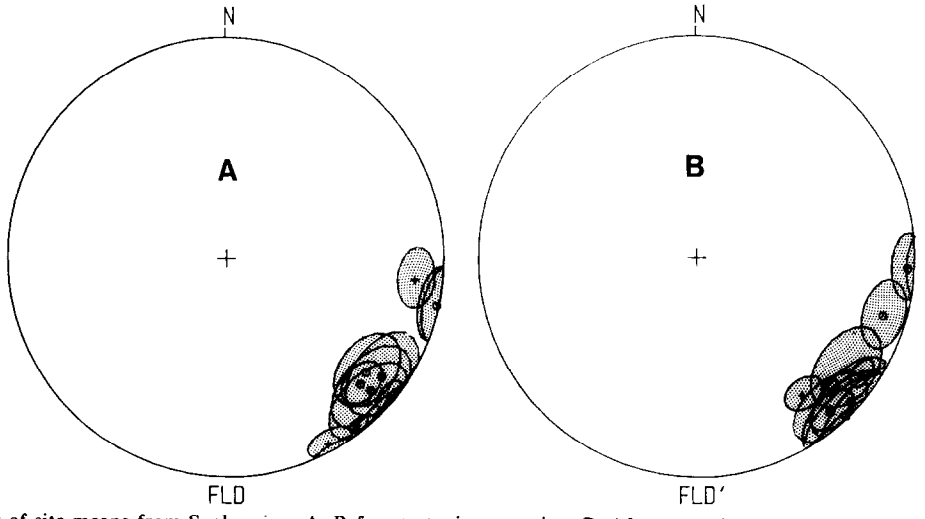


Fig. 5. Summary of site means from Sual region. A. Before tectonic correction. B. After tectonic correction. Coventions; equal area plots, + lower hemisphere, O upper hemisphere, with α_{95} intervals.

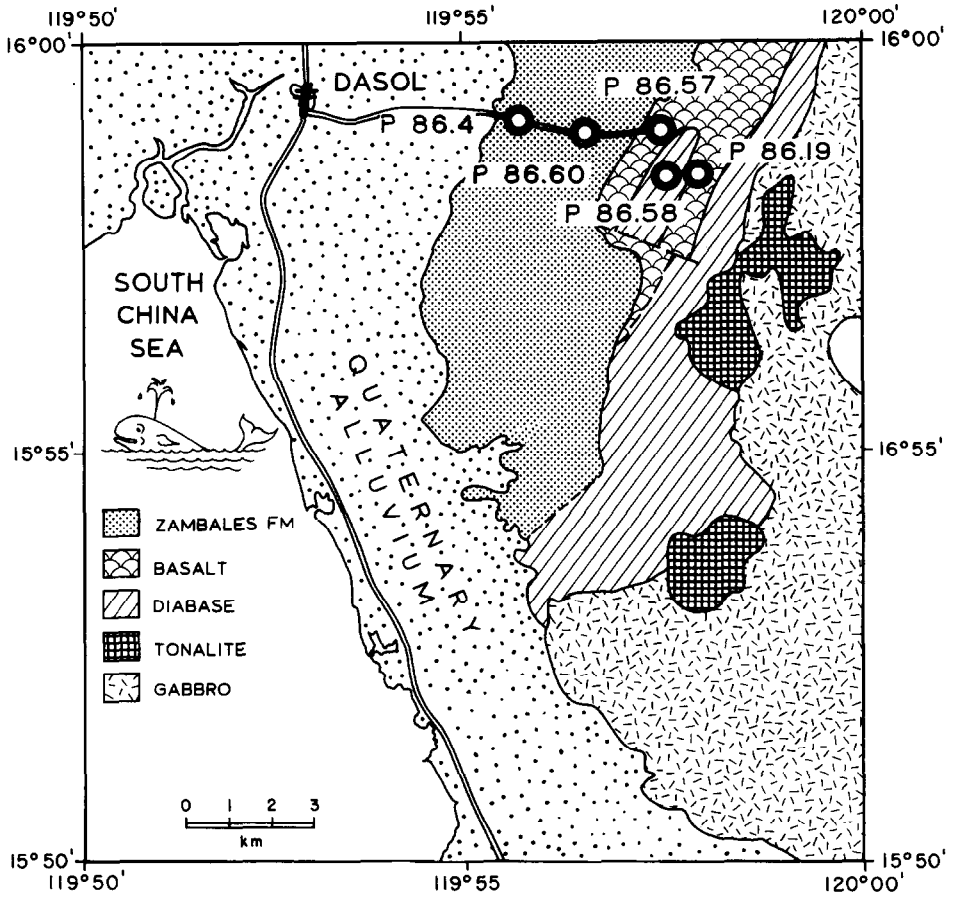


Fig. 6. Map of Barlo region sites (geology from Philippine Bureau of Mines and Geo-sciences—Dasol quadrangle map).

them, is primary. Both directions are defined by samples showing good endpoints characteristic of the isolation of single stable component of magnetization. The data define a strong easterly declination anomaly. The term declination anomaly is used here and subsequently to describe directions clearly departing from the present field, whether caused by clockwise, or anticlockwise rotations. Inclination is consistently shallow indicating equatorial latitudes.

Barlo Mine sites

The Barlo Mine is in the Dasol quadrangle. The uppermost units of the Acoje block of the ophiolite are seen here. Massive sulphides were mined in the pillow basalts of the ophiolite, presumably at former hydrothermal vent systems. Together with the pelagic sediments immediately above them, the flows make up the chert–spilite formation. A small sill was found in the pillow lavas and yielded a whole rock K/Ar age of 44.1 ± 3.0 Ma, demonstrating that the ophiolite section is no younger than Late Eocene and comparable in age to the intrusive near Sual. Above the chert–spilite formation lie the gently dipping Miocene sandstones of the Zambales Formation and Late Miocene reef limestone. All of the units have been sampled (Fig. 6).

Results

No consistent directions of magnetization were found in the reef limestone apart from a present field overprint. Three sites in the Miocene sandstones gave internally consistent directions. Most samples have a single component of magnetization revealed by either thermal, or AF demagnetization. AF demagnetization reveals that the magnetization is soft, while thermal demagnetization reveals that it has distributed blocking temperatures.

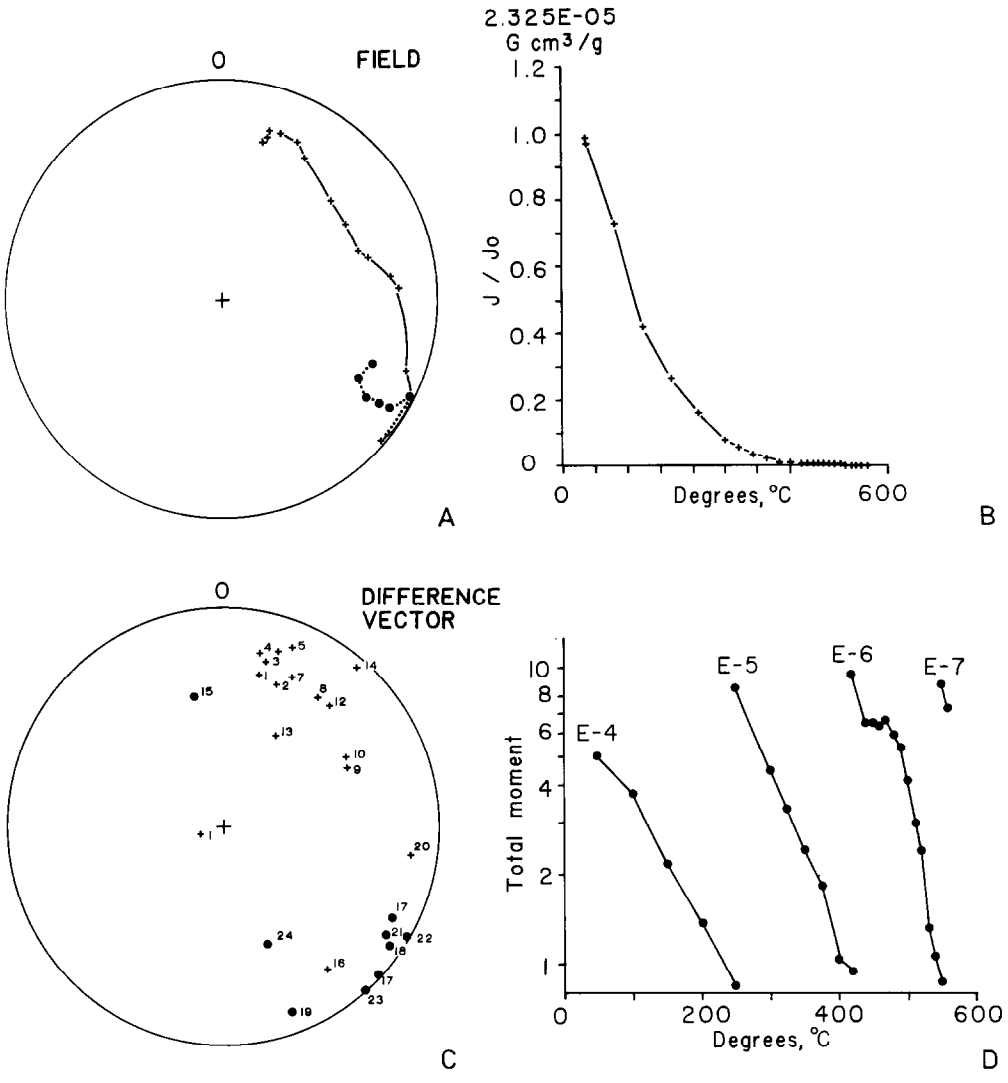


Fig. 7. Thermal demagnetization with measurements at high temperature of the Zambales Formation. A. Equal area plot of vector directions. B. Plot of intensity. C. Equal area plot of difference vectors. D. Logarithmic plot of intensity.

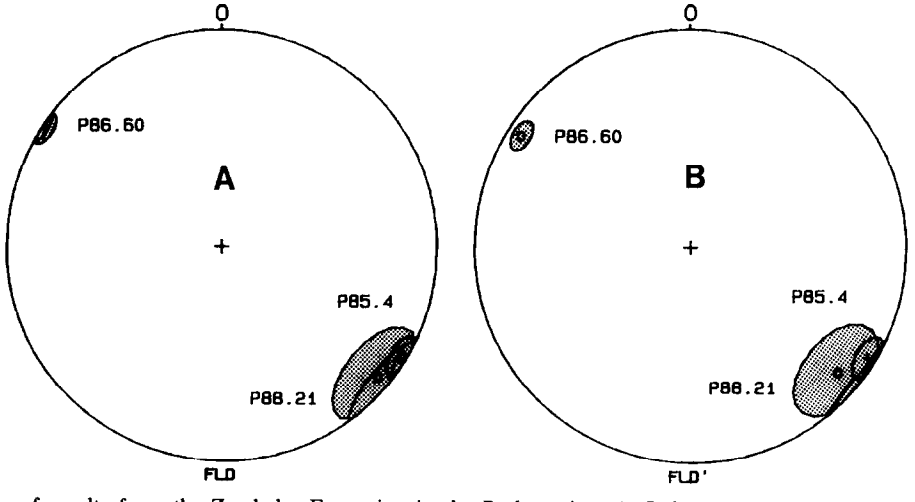


Fig. 8. Summary of results from the Zambales Formation in the Barlo region. A. Before tectonic correction. B. After tectonic correction.

TABLE 2

Summary of results from Barlo

Site No.	Unit	Location lat. (°)(′)	Location long. (°) (′)	N	Demag. AF/TH	Inc. (°)	Dec. (°)	R	k	α_{95}	Struct. corr.	Inc′. (°)	Dec′. (°)
Early to Mid-Miocene													
85.4	Zam. Fm.	15 59	119 56	4	T300–500	−1.8	121.4	3.99	206.7	6.4	150/12S	+3.9	121.6
86.00	Zam. Fm.	15 59	119 57	8	AF/TH SC	+1.4	303.3	7.95	145.7	4.6	225/9W	−7.4	303.2
88.21	Zam. Fm.	16 05	119 52	4	T300–500	−15.5	130.5	3.9	32.1	16.4	195/10W	−6.4	129.7
Oligocene-Eocene: chert-spilite													
86.58	sed.	15 58	119 57	12	T300–500	−21.0	122.5	11.64	30.9	7.9	210/35W	+14.0	122.5
79.1	sed.	15 58	119 56	5	A200–400	+14.3	85.6	4.99	378.2	3.9	210/10W	+22.5	083.7
86.57	sill.	15 58	119 57	5	T300–500	−35.3	36.4	4.00	571.0	3.8	210/35W	−24.9	065.6

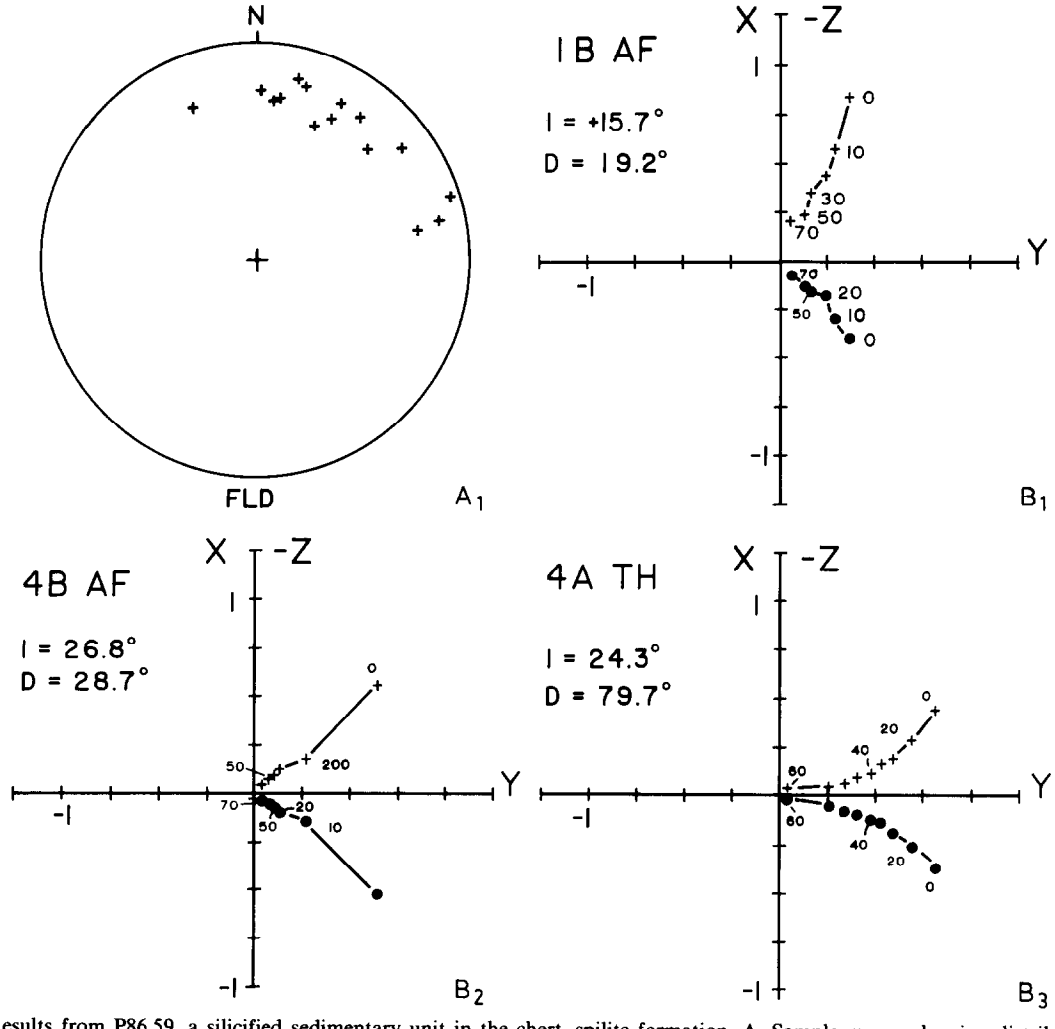


Fig. 9. Results from P86.59, a silicified sedimentary unit in the chert–spilite formation. A. Sample means showing distribution of stable directions between northerly and easterly azimuths. B. Zijderveld plots for AF and thermal demagnetization of sample P86-59.1 and 4. Conventions; Zijderveld orthogonal plots are in field coordinates, i.e. $+X$ is north, $+Y$ east and $+Z$ is vertically down. The XY plots are represented as + and the ZY as O. Demagnetization values are given in mT for AF and °C for thermal.

At all three sites, the magnetization is shallowly inclined. At site P86.60, it has a northwesterly declination and at P85.4, it is antipodal, satisfying a reversal test. At P88.21, two declinations are seen, one northeasterly and a second similar to that seen at P85.4. Both are seen in a single sample during thermal demagnetization with the northeasterly direction being blocked at lower temperature (Fig. 7). Zambales formation results are given in Fig. 8 and Table 2.

Four sites were collected in the sediments of the chert spilite group. P86.58 gave a similar direction to that seen in the Miocene Zambales Formation. Two sites, from lower in the section exhibited marked silicification. The first P86.59 revealed a strong present field overprint. Some samples yielded a single direction of magnetization upon thermal and AF demagnetization close to the present field. Others gave single directions at various distance from the present field direction with convergent Zijderveld diagrams. In some of the plots, the vector begins to retrace its path along the great circle back towards the present field. This is diagnostic of overlapping blocking or AF spectra, which preclude isolation of a single component. Note however, that the Zijderveld plots give indications of convergence. A plot of the dominant direction in each sample is given in Fig. 9, with representative Zijderveld plots. Samples from site P79.1, which is at the base of the sedimentary section, exhibited similar behavior, although all samples showed substantial movement away from the present field during demagnetization to give more consistently easterly directions. Site P86.19 is from the east side of the mine pit and includes basalts and intermixed chert. Two directions emerged from thermal demagnetization. Both are only poorly defined. One has a declination of about 50° and the other 150°. Both basalts and cherts and found with each direction. There is also clear evidence of contamination in an unrotated reverse field direction, which we have been unable to eliminate.

A single small sill (P86.57) was sampled in the pillow lavas and gave a well defined northeasterly azimuth with a negative inclination. All samples give excellent convergence on Zijderveld orthogonal diagrams and indeed in most samples the magnetization is single component. In all samples, NRM is predominantly blocked at high temperature. Figure 10 summarizes the results from the chert–spilite sites, for which the data merited Fisher analysis. The dominant directions again define an easterly declination anomaly.

NRM : IRM (s) plots have been made for the principal rock types encountered in this region. The results from the sill yield an initial ratio of 10^{-2} with concave down curves, suggesting the presence of primary NRM—a TRM in fine grain carriers. The flow from site P86.19 gives a similar concave down curve but a lower initial ratio, suggesting the presence of fine carriers, but not of

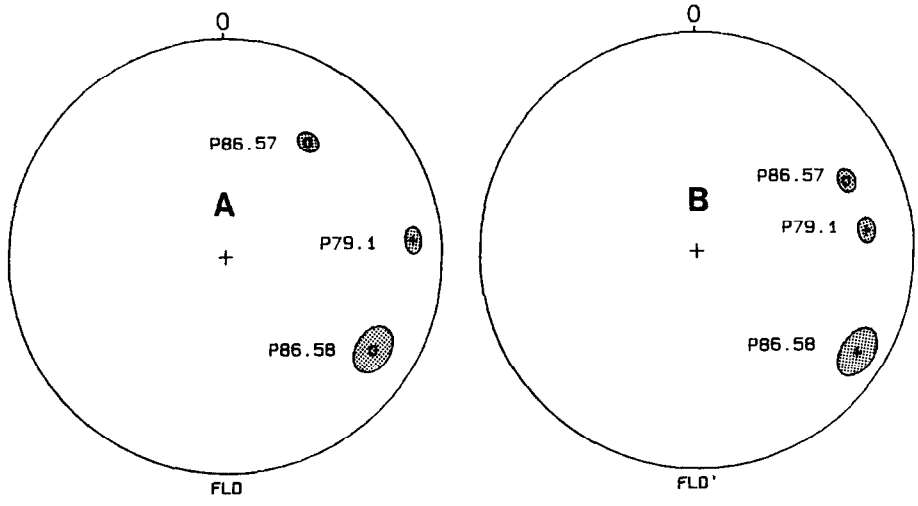


Fig. 10. Summary of site means from chert–spilite from Barlo. A. Before tectonic correction. B. After tectonic correction.

a primary TRM. The Miocene sediment at site P86.60 plots in a range more typical of igneous rocks than sediments. It has the highest ratio of NRM : IRM (s) observed. The magnetization is largely carried by very soft multidomain material. The claystone, P86.58.20 from the chert–spilite group has a slightly larger IRM (s), but a much smaller NRM than the Miocene sample, giving a very low NRM : IRM (s) ratio. The chert from site P86.19 exhibits intermediate values, more typical of DRM or PDRM, although the absence of control data for sediments reduces the diagnostic value of the diagrams.

Interpretation

The sites in the Zambales Formation yield antipodal directions passing a reversal test, but the tectonic corrections are so small that no fold test is possible. These directions are Middle Miocene, or younger, so that the low inclination is puzzling. The α_{95} confidence intervals of these directions include the mean direction of P86.58 in the underlying chert spilite sites. The remaining site for which Fisher statistics could be computed, P79.1, gives an easterly declination with shallow positive inclination. They are distinguishable from the Miocene directions, although the presence of a strong normal overprint makes isolation of a final uncontaminated direction difficult. The sill P86.57 has single component magnetization, which upon tectonic correction, approaches the direction seen at P79.1. A similar direction is one of the two poorly defined directions in the flows and cherts of site P86.19.

The tectonic corrections for the Miocene sediments in Barlo are so small that they do not change the direction significantly. A westerly declination anomaly of about 60° is defined for these Miocene sediments. The same direction is seen in one of the Oligocene sediment sites, suggesting that this sediment was remagnetized during the Miocene. The other Oligocene site gives a declination of 83.7°.

The difference in direction between the Miocene and Oligocene magnetizations is accounted for by about 40° of counterclockwise rotation, or 140° of clockwise. The more plausible rotation is the counterclockwise. This is then consistent with the Oligocene sediments being reversely magnetized and exhibiting an additional counterclockwise rotation prior to the time of magnetization of the Miocene samples.

The result from the single sill is strikingly different from the other directions observed. According to its demagnetization characteristics, it is also very likely to represent a primary direction of magnetization. Given that its age is 44.1 ± 3.0 Ma, the simplest interpretation is that its NRM records the initial sea-floor magnetization of the ophiolite. Before tectonic correction its direction is inclined upward to the northeast. This direction cannot be a late remagnetization recording clockwise rotation because Zambales would have to be in the southern hemisphere. Upon tectonic correction the NRM approaches the direction of site P79.1 suggesting an additional counterclockwise rotation of about 20° between Eocene and the age of the Oligocene sediments, giving a net counterclockwise rotation of 115°. However, a single determination from a small sill does not average secular variation.

Acoje Mine and southern outcrops of the Acoje block

At the Acoje Mine in the Campbert quadrangle, the deeper parts of the ophiolite are exposed and the lower gabbros and layered dunite were sampled. The Acoje block extends well to the south of the mine being bounded by the Lawis fault, near to the Coto Mine in the Benguet Mines quadrangle, some tens of kilometers to the south (Fig. 11). The results from the gabbros and small dykes in the gabbros alongside the access road to the Coto Mine are therefore included in this group of sites. The results are described in ascending stratigraphic order within the ophiolite.

Results

Site P79.16 in the layered dunite at the base of the layered section gave consistent results after AF demagnetization from which the site mean declination of 287° and inclination of −29.0° with the α_{95} of 14.4° was calculated. Three samples did not yield a stable endpoint, but did upon thermal demagnetization.

The gabbros at P79.15 immediately above the dunite also gave a predominant direction with a

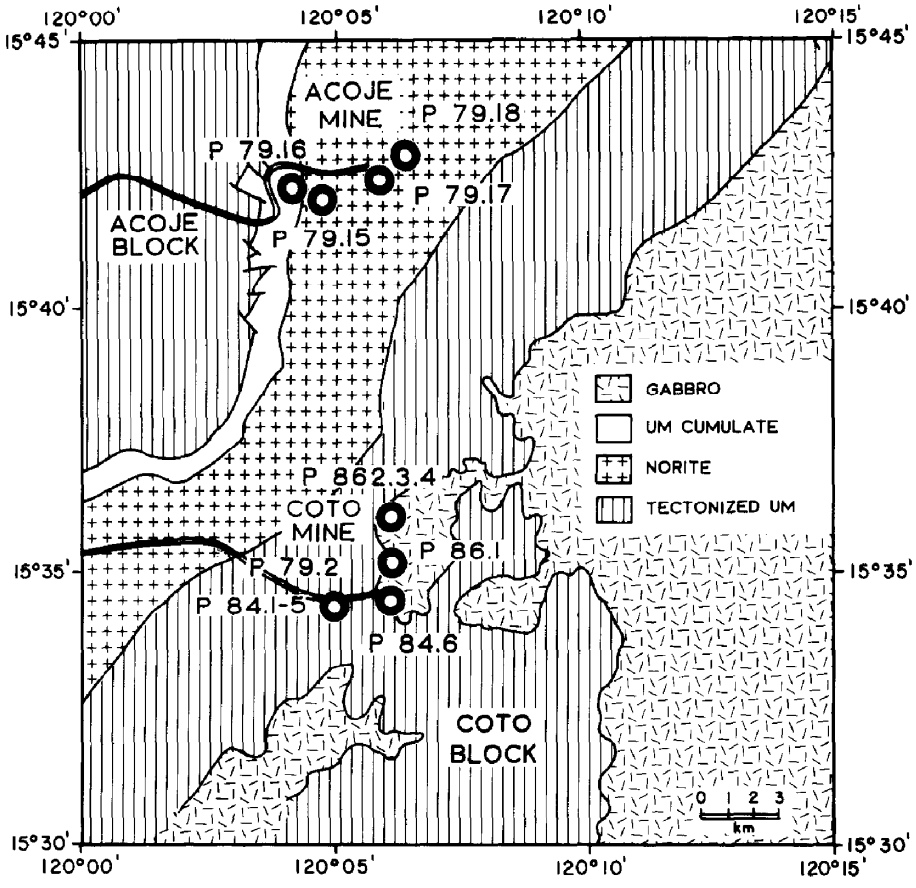


Fig. 11. Map of sites in the Acoje and Coto Mine area (geology from Hawkins and Evans, 1983).

few sample giving erratic results. These samples were not included in the Fisher statistics. The predominant direction had a declination of 102.2° and an inclination of +16.1°, with an α_{95} of 4.7°. In view of the inconsistent results from this site, a nearby site was collected in 1984 as P84.1. Con-

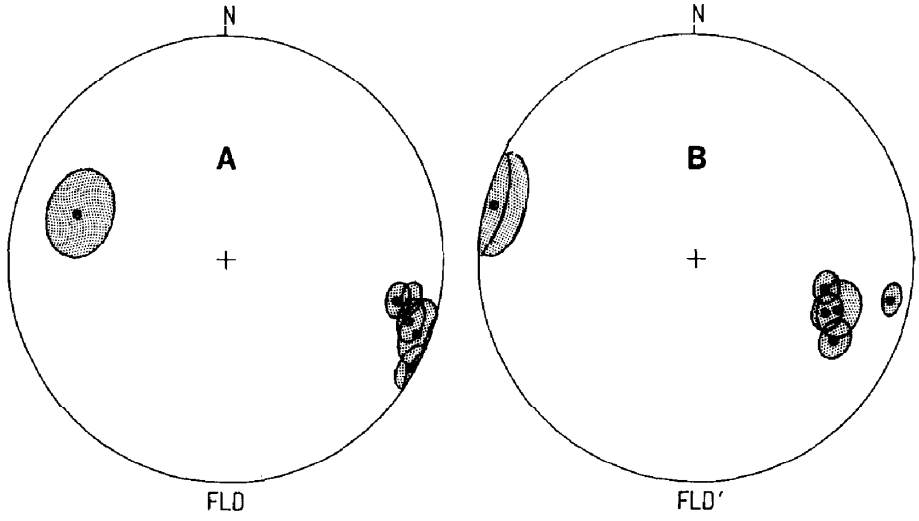


Fig. 12. Summary of site means from the Acoje Mine and Coto access road of the Acoje Block. A. Before tectonic correction. B. After tectonic correction.

TABLE 3

Summary of results from Acoje Mine and Coto access road of the Acoje block

Site No.	Unit	Location lat. (°)(′)	Location long. (°)(′)	N	Demag. AF/TH	Inc. (°)	Dec. (°)	R	k	α_{95}	Struct corr.	Inc′. (°)	Dec′. (°)
Eocene.													
Acoje Mine													
79.18	Gabbro	15 43	120 06	12	A/T SC	−12.3	108.3	11.78	50.2	6.2	000/25E	−35.8	112.2
79.15	Gabbro	15 42	120 05	12	A/T SC	14.7	101.9	11.9	109.4	4.7	000/25E	−9.8	101.7
84.1	Gabbro	15 42	120 05	11	TH	−20.5	103.5	10.9	93.2	5.5	000/25E	−3.9	102.7
79.16	Dunite	15 42	120 04	8	A/T SC	−29.0	287.0	7.56	15.8	14.4	000/25E	−4.9	284.9
Coto Approach Rd.													
79.41	Dyke	15 35	120 00	5	A/T SC	−6.7	110.8	4.94	70.8	9.2	030/25E	−31.3	109.3
79.40	Gabbro	15 35	120 00	12	A/T SC	−2.8	120.0	11.76	45.5	6.5	030/25E	−27.8	120.0

sistent results were obtained, giving a declination of 103.5° and inclination of +20.5°, close to antipodal to the dunite site.

A second site in the gabbros gave inconsistent results which were distributed along a great circle between directions with a declination of 300° and an inclination of −2° and a direction with declination of 212° and inclination of +17°. No Fisher statistics were calculated for this site.

P79.18, the final site in the gabbros at Acoje Mine gave roughly antipodal directions with declinations and inclinations of 278° and +7° and 108° and −12.3°. The two directions came systematically from samples from two different parts of the site, defining two sub-sites.

The site on the access road to the Coto Mine yielded results from gabbros and small dykes that intrude them. The two directions are statistically indistinguishable at the 95% confidence level.

The results are summarized in Fig. 12 and Table 3. It is evident that there is consistency in the directions of magnetization between these sites which are separated by tens of kms.

NRM : IRM (s) analysis revealed a wide range in both NRM and IRM (s) values. The range in IRM (s) values reflects the different amounts of magnetite in anorthosites and the gabbros. The ratio NRM : IRM (s) is less variable. The form of the demagnetization curves reveals a mixture of very coarse and very fine carriers and is consistent with a primary NRM (Fuller et al., 1988).

Interpretation

The internal consistency between these sites suggests that the area is not badly distorted by local tectonic effects. The occurrence of normal and reversely magnetized sites demonstrates that the magnetization could not have been acquired at a single time. The weakly negative directions to the southeast are evidence of later remagnetization, being similar to directions in overlying sediments. Tectonic corrections this deep in the body are clearly highly speculative. If one obtains a strike from the mapped contacts and uses a dip from the east side of the body, one then finds that the tectonically corrected directions are shallow. If one interprets a correction from the structure proposed by Rossman (1964), the correction would be much smaller leaving the direction essentially unchanged. This gives shallower inclinations, than obtained with the other tectonic correction but makes little difference to declination values.

Coto Mine

At the Coto Mine in the Benguet Mine quadrangle, there are exposures of tectonized peridotite, gabbros and plentiful dykes. These are diabasic dykes cutting saxonite and more rarely the gabbros above. We have collected from the gabbros and from the dykes, both where they intrude the saxonite and the grabbros. One of the dykes cutting the gabbros gave the whole rock

TABLE 4

Summary of results from Coto Mine

Site No.	Unit	Location lat. (°)(′)	Location long. (°) (′)	N	Demag. AF/TH	Inc. (°)	Dec. (°)	R	k	α_{95}	Struct. corr.	Inc′. (°)	Dec′. (°)
Coto Mine upper section													
86.1.3	dyke	15 35	120 06	10	A/T SC	+22.0	051.4	9.11	78.3	5.9	030/25E	11.3	058.3
86.1.1	cont 1	15 35	120 06	5	A/T SC	+9.5	061.3	4.89	38.0	12.6	030/25E	−3.8	062.4
86.1.2	incl.	15 35	120 06	5	A/T SC	+11.9	042.5	4.93	56.2	10.3	030/25E	5.6	046.3
86.2	gabbro	15 35	120 06	5	A/T SC	+11.3	038.7	4.96	90.6	8.1	030/25E	6.6	042.6
86.3	dyke	15 35	120 06	4	A/T SC	+8.8	044.7	3.65	8.6	33.3	030/25E	1.9	047.0
86.4	gabbro	15 35	120 06	8	A/T SC	+5.2	046.3	7.99	533.8	2.4	030/25E	−2.1	047.0
Coto Mine 900 level													
79.2	dyke	15 33	120 05	9	A/T SC	+7.7	058.9	8.65	23.1	10.9	030/25E	−4.4	059.7
84.1	dyke	15 33	120 05	10	A/T SC	−16.1	078.8	9.82	48.8	7.0	030/25E	−31.8	070.6
84.2	dyke	15 33	120 05	12	A/T SC	−21.5	064.3	11.94	189.9	3.2	030/25E	−21.6	062.8
84.3	dyke	15 35	120 05	5	A/T SC	−13.7	053.6	4.95	85.9	8.3	030/25E	−22.3	045.8
84.4	dyke	15 35	120 05	6	A/T SC	−7.8	063.7	4.98	164.7	5.2	030/25E	−20.8	058.1
84.5	dyke	15 35	120 05	9	A/T SC	−17.3	059.1	8.88	65.2	7.5	030/25E	−27.8	049.5
Site in gabbros													
84.6	gabbro	15 35	120 06	12	A/T SC	+24.7	056.7	11.63	29.4	8.1	030/25E	8.1	070.2

K/Ar age of 46.6 ± 5.1 Ma, indicating that the dykes are associated with the Eocene ophiolite and not with a later intrusive event.

Results

The results from the dykes are separated into two groups, those from the dykes in the saxonite and those in the gabbros. The upper dykes were sampled at two sites and an attempt made to carry out a baked contact test. At site P86.1, samples from the baked gabbro contact, the dykes and included gabbro xenoliths all gave statistically indistinguishable results. At the second site, the magnetization of the dyke was poorly defined and indistinguishable from the gabbro away from the dyke.

The lower dykes yielded stable end points in AF and thermal demagnetization. They gave a set of directions predominantly shallowly inclined upward to the east-northeast.

P79.21 is near the base of the gabbros and gave inconsistent results, which has shallow inclinations, but spread in azimuth from north of east to south. No Fisher statistics could be calculated for the site. A subsequent collection from nearby gave the results listed under P84.6. These are similar to the dykes, but more steeply positively inclined.

The paleomagnetic results are summarized in Table 4 and in Fig. 13. There are two separate groups of directions, but an overall mean defines a shallow inclination and a northeasterly declination.

The NRM : IRM (s) plot for the dykes in the upper gabbros yielded a high initial ratio and demagnetization characteristics typical of fine grain particles carrying weak field TRM. Similar results came from the lower dykes. Gabbros into which the upper dykes are intruded gave lower NRM : IRM (s) ratios and less obviously concave down curves.

Interpretation

The dykes yield two internally consistent sets of directions. The NRM : IRM (s) plots suggest that both groups carry stable weak field thermal remanence. Their NRM direction is different from all other directions found in Zambales with the exception of the sill from Barlo Mine. The uncorrected direction for the lower group is northeasterly and shallowly inclined upward, which is un-

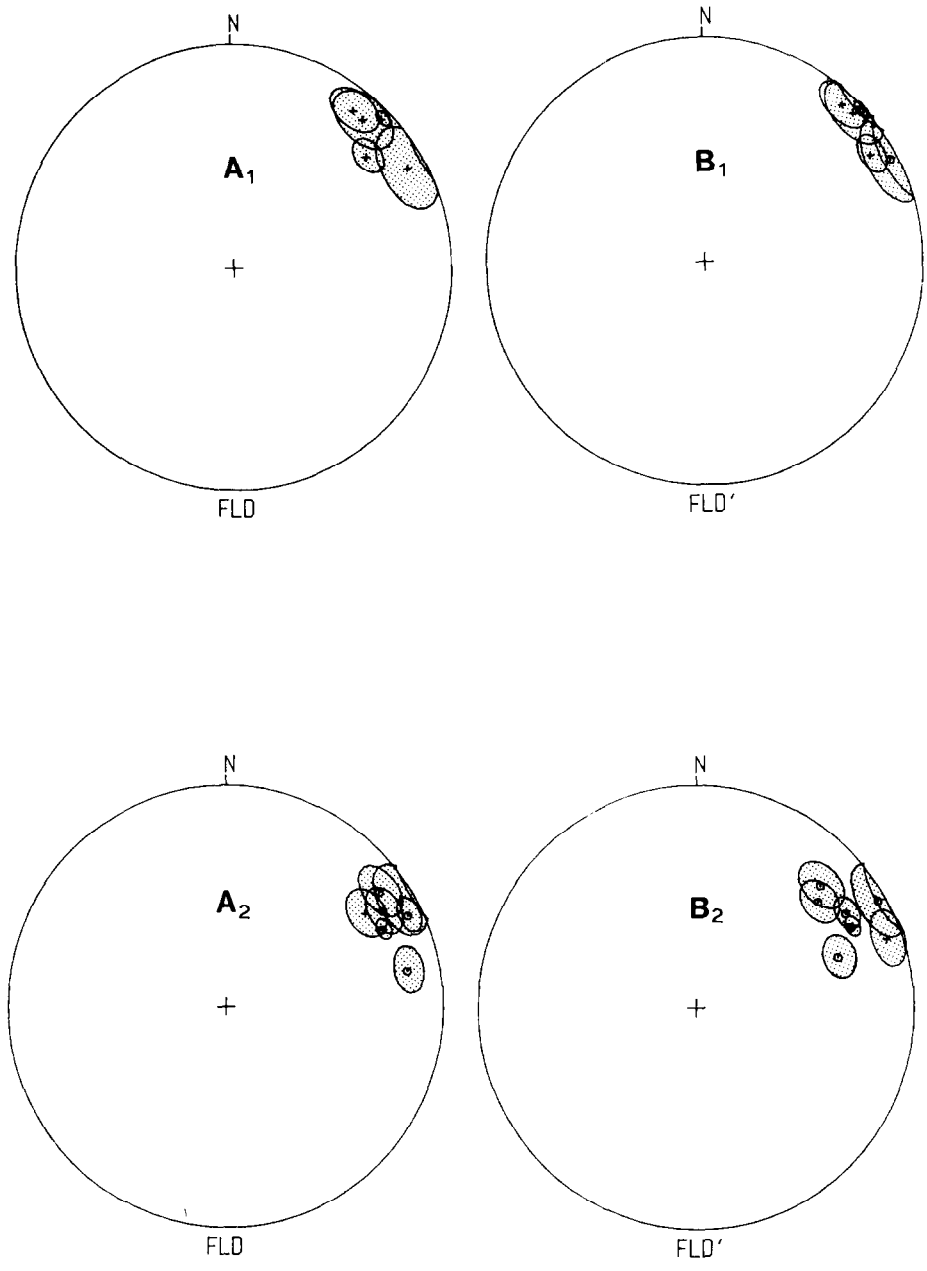


Fig. 13. Summary of site means from the Coto Mine of the Coto Block. A_1, B_1. Lower dykes. A_2, B_2. Upper dykes. A. Before tectonic correction. B. After tectonic correction.

likely to be a late remagnetization recording clockwise rotation because this would require Zambales to be in the southern hemisphere, when this late remagnetization was acquired. We therefore suggest that the dykes are recording an early sea-floor magnetization of the ophiolite, which was not reset by later events because of its high blocking temperature.

The tectonic correction for the dykes and the gabbro at the Coto Mine is highly speculative as it is at the Acoje Mine. Any correction based upon the general easterly dip of the ophiolite will make

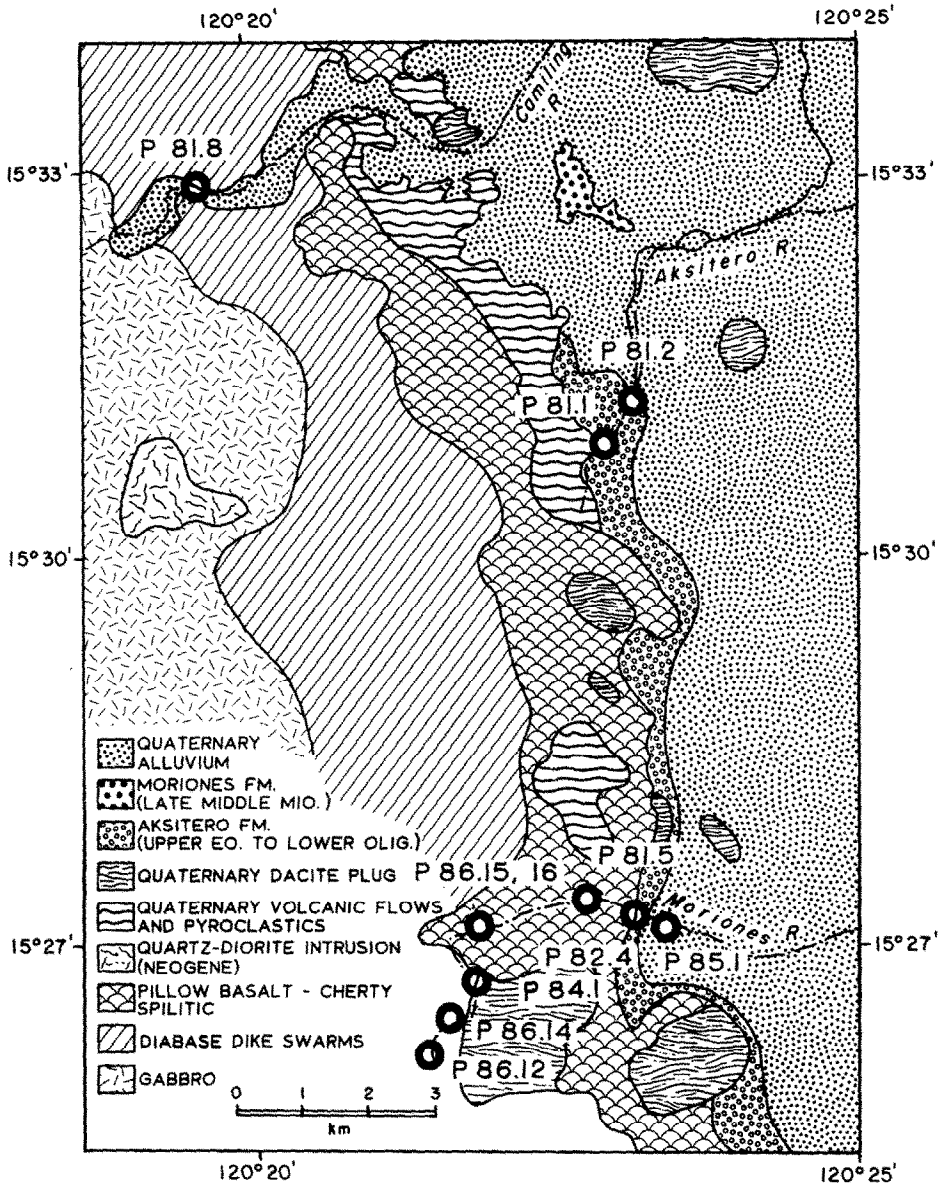


Fig. 14. Map of sites in the Aksitero, Moriones and Camelling rivers (Geology Philippine Bureau of Mines and Geo-sciences—San Ignacio and O'Donnel quadrangles).

the shallow inclination observed more negative. Using a correction based upon the structure interpreted by Rossman (1964) would leave the directions essentially unchanged. In either case, the inclinations remain shallow.

River sections on the east side of the Coto block

The overall dip of the central part of the Zambales range is easterly so that the sediments and uppermost parts of the ophiolite itself are seen on the east side (Fig. 14). Several river sections in the San Ignacio and O'Donnel quadrangles in Tarlac have been sampled to determine the paleomagnetism of these units. In this area there are also younger intrusions, of Late Miocene, or Pliocene age, which have also been studied paleomagnetically by McCabe et al. (1987).

Results

One of the young intrusions was sampled. It gave a mean of declination of 193.4° and inclina-

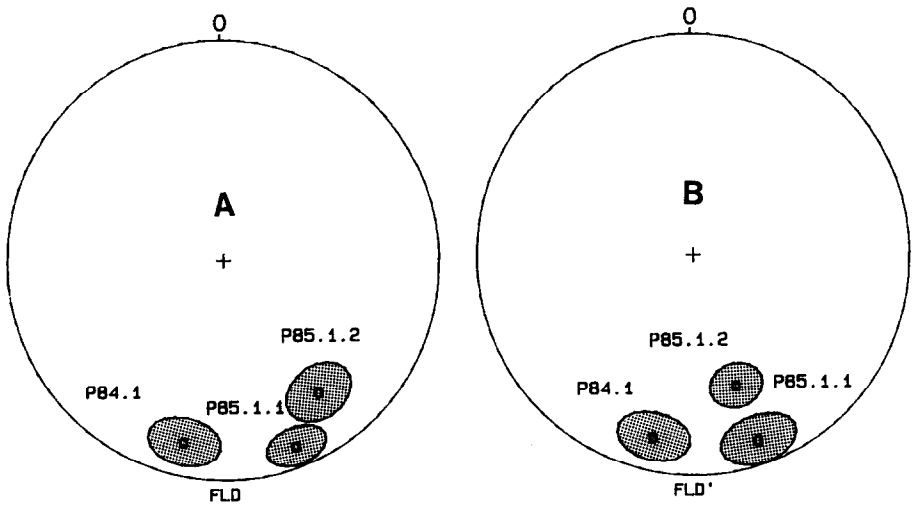


Fig. 15. Summary of site means for the Pliocene–Late Miocene intrusion and Moriones Fm. from the Moriones River. A. Before tectonic correction. B. After tectonic correction.

TABLE 5

Summary of results from river sections in Tarlac

Site No.	Unit	Location lat. (°)(′)	Location long. (°) (′)	N	Demag. AF/TH	Inc. (°)	Dec. (°)	R	k	α_{95}	Struct. corr.	Inc′. (°)	Dec′. (°)
Late Miocene–Pliocene													
84.1	plug.	15 27	120 22	7	A/T SC	−16.3	193.4	6.80	29.9	11.2	–	−16.3	193.4
Early Miocene													
85.1.1	Mor. Fm.	15 27	120 24	6	T450–500	−10.1	159.1	5.91	55.1	9.1	040/42S	−37.7	162.3
85.1.2	Mor. Fm.	15 27	120 24	8	T450–500	−26.7	144.4	7.71	24.0	11.5	305/20N	−11.8	161.0
Eocene–Oligocene													
81.1.1	Aks. Fm.	15 31	120 23	8	T450–550	−15.2	300.1	7.51	14.2	15.2	340/25N	−2.0	297.8
81.1.2	Aks. Fm.	15 31	120 23	10	T450–550	14.6	083.4	9.53	19.5	11.2	340/25N	−4.9	083.0
81.1.R	Aks. Fm.	15 31	120 23	6	AF/T SC	27.4	016.0	5.80	33.3	11.8			
82.4.1	Aks. Fm.	15 27	120 23	4	T450–550	−26.8	144.6	2.99	308.3	7.0	340/25N	−30.6	158.4
82.4.2	Aks. Fm.	15 27	120 23	4	T450–550	−6.5	286.0	3.98	182.4	6.8	340/25N	13.7	287.0
82.4.3	Aks. Fm.	15 27	120 23	3	T450–550	17.5	113.7	2.89	18.7	29.3	340/25N	−1.1	111.2
82.4.5	Aks. Fm.	15 27	120 23	5	T450–550	−17.0	264.9	4.95	87.1	8.2	340/25N	7.2	264.4
82.4.R	Aks. Fm.	15 27	120 23	6	AF/T SC	11.6	027.1	5.70	16.9	16.8	–	11.6	027.1
Ophiolite													
81.8	sill.	15 33	120 19	8	A/T SC	42.8	110.7	7.92	85.4	6.0	340/25N	22.4	101.2
81.5	sill.	15 27	120 23	6	T AF SC	2.2	100.9	5.95	100.8	6.7	340/25N	−19.1	102.9
86.15	sill.	15 27	120 22	4	A/T SC	30.5	103.0	3.94	48.8	13.3	340/25N	8.9	098.4
86.16	sill.	15 27	120 22	4	A/T SC	31.6	110.7	3.95	61.4	11.8	340/25N	11.7	104.5
86.13.1	dyke	15 26	120 21	3	A/T SC	24.1	078.5	2.98	97.1	12.6	340/25N	−0.7	077.8
86.13.2	cont.	15 26	120 21	6	A/T SC	7.2	097.9	5.97	177.9	5.0	340/25N	−14.9	098.7
86.12	gabbro	15 26	120 21	5	A/T SC	+13.8	112.2	4.99	462.8	3.6	340/25N	−5.0	110.9
86.14	gabbro	15 26	120 21	5	A/T SC	+15.2	110.1	4.98	184.6	5.6	340/25N	−4.3	108.6

tion of $-16.3°$ with an α_{95} of 11.9° and is thus marginally distinguishable from the unrotated reversed field direction. The results are presented in Fig. 15 and Table 5.

A large collection was made from the Moriones Formation in the Moriones River. For the most part, it is very coarse and not suitable for paleomagnetism. However, collections were made in finer units and in particular in a small plunging fold. The directions gave westerly declination anomalies (Fig. 16), but the combination of the magnitude of the scatter and the small dips of the beds on either limb of the fold precluded a satisfactory fold test. P79.13 from near to the base of the formation yield northwesterly declinations, but are partially overprinted. A third site, in a fine shale higher in the section yielded present reversed field directions and some samples with a southwesterly declination. The third site is not included in the figure and table because consistent directions were not obtained.

The Eocene to Mid-Oligocene pelagic sediments of the Aksitero Formation overlie the flows of the Ophiolite. In the type section, individual samples were taken at precise stratigraphic heights, with substantial collections at a number of these sites. The lowermost members have low intensities of NRM of order 10^{-7} A m^2 kg^{-1}. From the base of the section to a stratigraphic height of about 15 m, the samples are predominantly magnetized to the northwest with shallow upward inclinations (Fig. 16). This is indistinguishable from the direction found in the Miocene Moriones Formation. Other samples were found to have present field directions and directions close to the reverse of the plug direction (Fig. 16). Individual samples revealed declinations of approximately 270°, which were distinguishable from the Moriones Formation direction. The only mean site direction not seen in younger rocks is that of P81.1.2 (Fig. 16). This direction with its declination of 83° which is found in samples from 20 to 30 m stratigraphic height in the section is of opposite polarity to the few samples in the lowermost sites which appear to have a pre-Miocene magnetization.

The section in the Moriones River is less than 30 m in stratigraphic height. It overlies the basalts of the ophiolite and is in turn overlain by the Moriones Formation. The lowermost site approximately 1.3 m above the basalts gives a declination of 150° and inclination of $-25°$ (Fig. 16), which is indistinguishable from the Moriones Formation direction. Above this site, P82.2 yields declinations to the northwest with shallow upward inclination. This is again indistinguishable from the Moriones Formation direction. The only site to give directions distinguishable from the Moriones Formation results is P82.5 with a mean declination of 264.9° and inclination of $-17.0°$ with an α_{95} of 8.2°. A prominent secondary direction seen throughout the section has a declination of 030° and an inclination of $+20°$. There is also

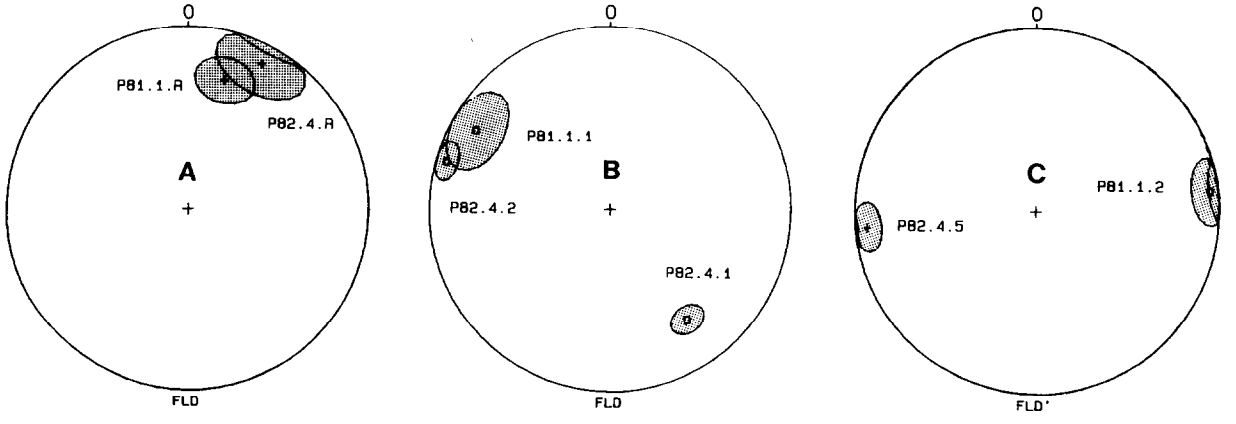


Fig. 16. NRM directions observed in the Eocene–Oligocene Aksitero formation from the Moriones and Aksitero River sections. A. Late Miocene–Pliocene secondary direction before tectonic correction. B. Middle–Early Miocene secondary direction before tectonic correction. C. Interpreted primary direction after tectonic correction.

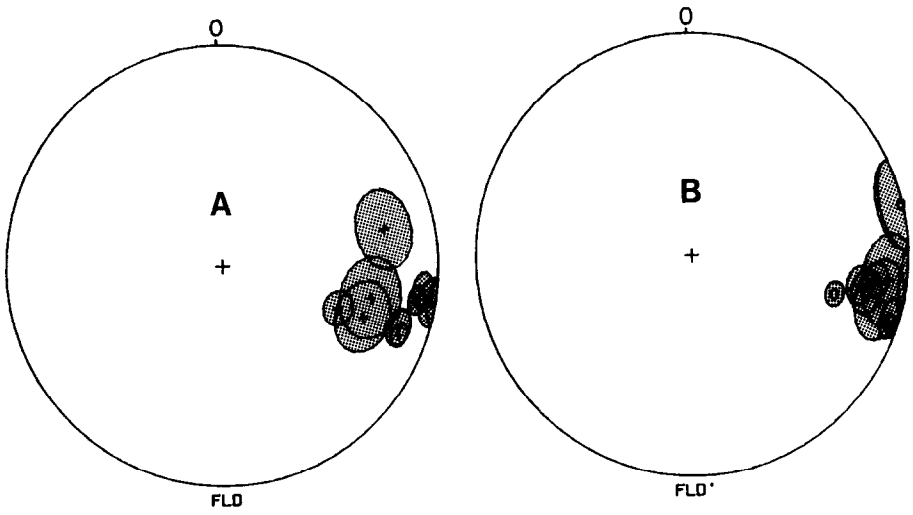


Fig. 17. Results from ophiolite in the Camelling and Moriones River sections. A. Before tectonic correction. B. After tectonic correction.

present field and unrotated reverse magnetization. The behavior is reminiscent of the sediments overlying the ophiolite in the north.

Sills in the Camelling and Moriones rivers gave good results with readily recognizable end points upon either AF or thermal demagnetization. The results for the four sites, two in each river are given in the summary table and Fig. 17.

The results from the gabbros come from the Moriones River and give consistent results with well defined endpoints whether AF, or thermal demagnetization is used. An attempt was made to carry out a baked contact test using a small dyke of about 1 m width which intrudes the gabbro. However, samples in the dyke and at varying distances from it up to a meter gave statistically indistinguishable results.

NRM : IRM (s) analysis for the gabbros yield results typical of weak field TRM in igneous rocks. In contrast, the sills of sites P85.15 and 16 have much weaker NRM and IRM (s) not typical of an igneous rock carrying TRM.

The sediments of the Aksitero Formation in the Moriones River section have low NRM and IRM (s). Sample P82.54 has very similar characteristics to those of dolomites of the Monterey Fm. of Santa Barbara County, California with primary remanence carried by fine detrital grains of magnetite (M. Hart, pers. commun., 1988). In contrast, P82.27 has behavior more like samples from the Monterey carrying secondary magnetization carried by authigenic magnetite (ibid.). The Moriones material has strong IRM (s) indicative of the presence of plentiful magnetite, but it is clearly soft.

Interpretation

The young intrusive from this region defines a small easterly declination anomaly. This is consistent with other results from the region (McCabe et al., 1987). Their very young age, which may be as little as 5 M.y. (E.E. Geary, pers. commun., 1987), ensures that this is indeed a small clockwise rotation.

In contrast to the young intrusives, the Early to Middle Miocene Moriones Formation shows a significantly larger westerly anomaly of between 30° and 40°. We interpret a counterclockwise sense of rotation between the time of magnetization of the Moriones and of the young intrusives because such a rotation is substantially smaller than the necessary clockwise rotation. The direction with a declination of approximately 20° and an inclination +30° is interpreted as a secondary direction generated by hydrothermal activity associated with the intrusives.

The results from the Eocene–Oligocene Aksitero Formation give a number of distinct directions, some of which can be identified as remagnetizations in directions seen in younger rocks. For example, the small easterly declination anomalies are interpreted as Late Miocene remag-

netization. Similarly the direction found at the lowermost site in the Moriones River (P82.4.1) is interpreted as an older remagnetization because it is the same as that seen in the Moriones Formation immediately above Aksitero. This is the direction shown by the sample with NRM : IRM (s) characteristics similar to the Monterey sample carrying secondary magnetization. The approximately E–W declinations are then interpreted as the most ancient magnetizations observed. The sample showing the characteristics similar to those carrying primary sample in the Monterey has such a direction, which is consistent with our interpretation.

The declination anomaly is interpreted to be caused by counterclockwise rotation because the difference between the Miocene Moriones Formation direction and that of the Aksitero Formation requires a small additional rotation in the sense, but a very large clockwise rotation.

The interpretation of counterclockwise rotation is consistent with the magnetostratigraphy. The Eocene–Oligocene boundary falls in the reversed part of chron 13, which lasts for a little more than a million years ending at 36 Ma. The detailed stratigraphy of the Aksitero (Schweller et al., 1983) predicts that there should be a polarity change from predominantly normal to predominantly reversed directions at a stratigraphic height of about 15 m above the base of the section. There is indeed a polarity change there from predominantly northeast directions below to predominantly southwest above. It therefore appears that the lower samples are indeed normal and the upper reversed, which is consistent with the proposed resolution of the polarity ambiguity.

In the ophiolite itself, a remarkably consistent set of results is seen. Before tectonic correction, they are similar to the results seen in the deep layered gabbros and the dunite of the Acoje block. After tectonic correction, using the overlying Aksitero Formation bedding, they give shallow inclinations and declinations ranging from approximately 080° to 110°. The time of magnetization of the sill complex of the ophiolite is not clear.

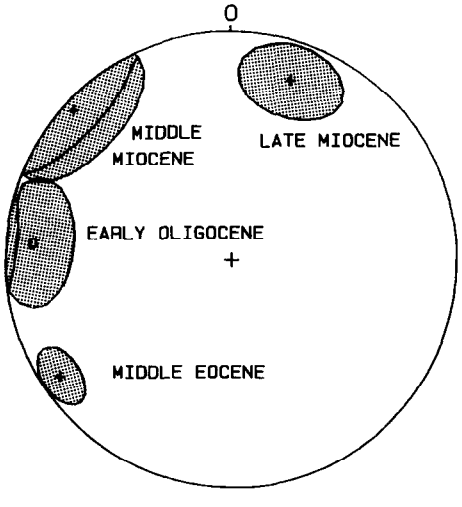


Fig. 18. Summary of paleomagnetic results from the Acoje and Coto blocks of the Zambales ophiolite. Interpreted normal polarity directions shown. (a) *Pliocene–Late Miocene;* plug direction and secondary magnetization in Aksitero; (b) *Middle Miocene:* Zambales and Moriones formations and secondary magnetization in Aksitero; (c) *Oligocene:* chert–spilite and Aksitero formations; (d) *Mid-Eocene*: sill from Barlo and dykes from the Coto Mine.

Summary

The results are discussed in chronological order starting with the youngest. It is an important aspect of our interpretation that, if the same direction is seen in units of different age, the magnetization in the older rock is secondary and acquired at the same time as the magnetization of the younger unit. The justification for this approach is that the tectonic activity of the region precludes long periods of time without movement with respect to the geomagnetic reference frame.

The results are illustrated in Fig. 18, in the form of summary plots of directions for the Acoje and Coto blocks. Results from the two blocks are plotted on a single diagram because our data do not suggest separate paleomagnetic histories for the two proposed blocks. In these figures the results are mapped into the interpreted normal polarity in the interest of clarity. The reader should recognize that this involves an element of interpretation not present in the earlier plots.

The youngest rocks studied were from one of the Pliocene–Late Miocene intrusions from the Tarlac region on the east side of the Coto block. The magnetization is marginally distinguishable

from an unrotated reversed field indicating that they may be slightly clockwise rotated. Secondary directions from the Aksitero are also included. In the figure the interpreted normal direction is plotted, as noted above.

Early to Mid-Miocene rocks were collected in the north from the Zambales Formation near Dasol and on the east side in the Moriones River section. The sites from the Zambales Formation exhibit westerly declination anomalies. The Mid-Miocene age of the formation is in this case only an upper limit on the age of the magnetization because there is no evidence that it was primary. A westerly declination anomaly was also found in the Moriones Formation in the Moriones River. The age of the sediment sets an upper limit on the age of the magnetization with the westerly declination anomaly. The sense of rotation which brought about the declination anomaly is interpreted to be counterclockwise.

The tectonic corrections are so small for these sediments that a convincing attitude test is not possible. In view of the short time available between Mid-Miocene and Pliocene, which is an upper limit on the age difference, a clockwise rotation of about 120° seems much less likely than the required 60° of counterclockwise rotation.

Eocene–Oligocene sediments were collected in the north, near Barlo, and in the south in the Aksitero and Moriones River sections. At Barlo two different directions were found. The one in a claystone overlying the chert–spilite is similar to those in the Miocene Zambales Formation and we interpret it as a secondary Miocene direction. The other direction is seen in the cherts directly overlying the basalts. It showed convergence on AF and thermal demagnetization Zijderveld plots. It has an easterly declination of almost 90° and shallow inclination. The direction is seen in the south in the Aksitero Formation with a reversal test. This direction is not seen in younger units.

The simplest interpretation of the Eocene–Oligocene result is that it represents additional counterclockwise rotation between the time of magnetization of the Miocene Zambales Formations and the time of magnetization of these Eocene–Oligocene chert–spilite sediments. A similar interpretation holds for the difference between the Aksitero and Moriones formation directions in the river sections in the southeast of the Coto block. This interpretation is consistent with the magnetostratigraphy of the Eocene–Oligocene boundary.

Turning to the rocks of the ophiolite, we see consistent results from the various units. The results from the uppermost units from Sual yield a range of declinations from southeast to east with shallow inclinations. The southeasterly direction is indistinguishable from the Miocene direction recorded in the Zambales Formation at Barlo and we therefore interpret it as a Miocene overprint. The more easterly directions are similar to those in the overlying Eocene–Oligocene sediments and may be closer to the primary magnetization of these units, but we are not confident that total separation of all secondary magnetization has been achieved. The sill at Barlo is interpreted as the primary direction for the ophiolite, but unfortunately a single small sill does not permit averaging of secular variation. Placing it within the framework of the rest of our interpretation, it represents a maximum counterclockwise rotation. In such a scenario, the less counterclockwise rotated directions are interpreted as unroofing magnetization, or magnetization associated with collisional driven processes, such as fluid migrations (e.g., Oliver, 1986).

The uppermost sills and gabbros were also sampled in the south near Tarlac. The results were distinguishable from the Miocene directions, having steeper positive inclinations. This direction is intermediate between that in the overlying Aksitero Formation and in the Moriones Formation and similar to that seen in the gabbros and sills in the Acoje block, suggesting that it was acquired during uplift and emplacement of the ophiolite.

The results from deep in the ophiolite yield two directions. The gabbros and dunite at the Acoje Mine and on the Coto Mine access road gave directions similar to those seen in the sills from the south in the Coto block at Tarlac. These demonstrate consistency of magnetization over a distance of tens of kilometers. The dykes give northeasterly declinations and shallow inclinations, similar to the direction in the sill from

Barlo. These magnetizations are blocked at very high temperatures and hence may have survived since the initial cooling of the sea floor.

The paleomagnetism of the Acoje and Coto blocks of the Zambales ophiolite and its sedimentary cover are consistent and suggest counterclockwise rotation and northward motion. The most rapid rotation was in the Miocene after much of the northward motion was completed. Since the Late Miocene, a small amount of clockwise rotation has taken place. There appear to have been two strong periods of secondary magnetization. The first is readily related to the igneous activity which gave rise to the Late Miocene–Early Pliocene plugs. The second, during Middle Miocene is more enigmatic and may be related to collision and emplacement of the ophiolite.

Comparison with other paleomagnetic studies of Luzon

The results and interpretations given in this paper are consistent with earlier work by Hsu (1972). We do not see the evidence for the very recent counterclockwise rotation described by De Boer et al. (1980). Our interpretation of counterclockwise rotation of Zambales is in conflict with a number of recent papers appearing with McCabe as senior author, e.g., McCabe et al. (1985, 1987), in which clockwise rotation of Luzon is suggested. The interpretation of the sense of the rotation of the region is so central to any tectonic applications of the results of the paleomagnetism, that we consider this discrepency, before any discussion of the tectonic implications of the paleomagnetism.

The evidence cited in support of clockwise rotation in the most recent paper (McCabe et al., 1987) is threefold:

(1) Six dykes from the Coto Mine are reported to show "declination values that are deflected 57° clockwise from the present field direction, suggesting that the region has undergone a strong post-intrusion clockwise rotation".

(2) An earlier result from Oligocene sediments reported in Fuller et al. (1983) indicates that "these sediments are rotated over 80° clockwise from the present dipole field direction".

(3) Preliminary results (R.J. McCabe, unpublished data, 1986) from four sites in the Oligocene portion of the Aksitero Formation "show evidence for around 50° of clockwise rotation" and include both normal and reversed samples.

These three points do not appear to us to be convincing evidence for clockwise rotation of Zambales. As discussed above, the first reason is based upon an incorrectly assigned age of the Coto dykes. The second does not set the critical direction of the Oligocene sediments in its stratigraphical setting. When this is done, the result is consistent with counterclockwise rotation. The third argument rests on unpublished evidence.

Taking the interpretation of counterclockwise rotation of the Zambales region, we now compare it with other results from the northern Philippines. To the north and east across the central Philippine basin, we have additional results from Baguio in Cordillera. Clockwise rotations are recorded back to at least about 10 Ma (Fuller, 1985). Prior to that counterclockwise rotation is interpreted there (Fuller et al., 1983). This has also been challenged by McCabe (1984). Nevertheless, as will be justified elsewhere, it is our interpretation that the same sense of rotation is recorded in both Zambales and the Cordillera back to the Early Miocene. In the Baguio region, the period of rapid counterclockwise rotation is between approximately 10 and 15 Ma. Additional work is underway to obtain data for older units in the Cordillera in the Baguio region.

To the south of Batangas, the Early to Mid-Miocene Talahib andesite flows give a westerly declination anomaly comparable to that seen in the Moriones Formation in the Moriones River, in the Zambales Formation and uppermost units of the ophiolite in the Zambales Formation and uppermost units of the ophiolite in the Barlo and Sual regions (Almasco and Fuller, 1986). This suggests that much of the western side of Luzon has a common tectonic history since Early–Mid-Miocene times. The Miocene was a period of rapid counterclockwise rotation followed from about 10 Ma by a clockwise rotation of some 10° to 20°.

Continuing to the south a very similar pattern of paleomagnetic results is seen in Marinduque.

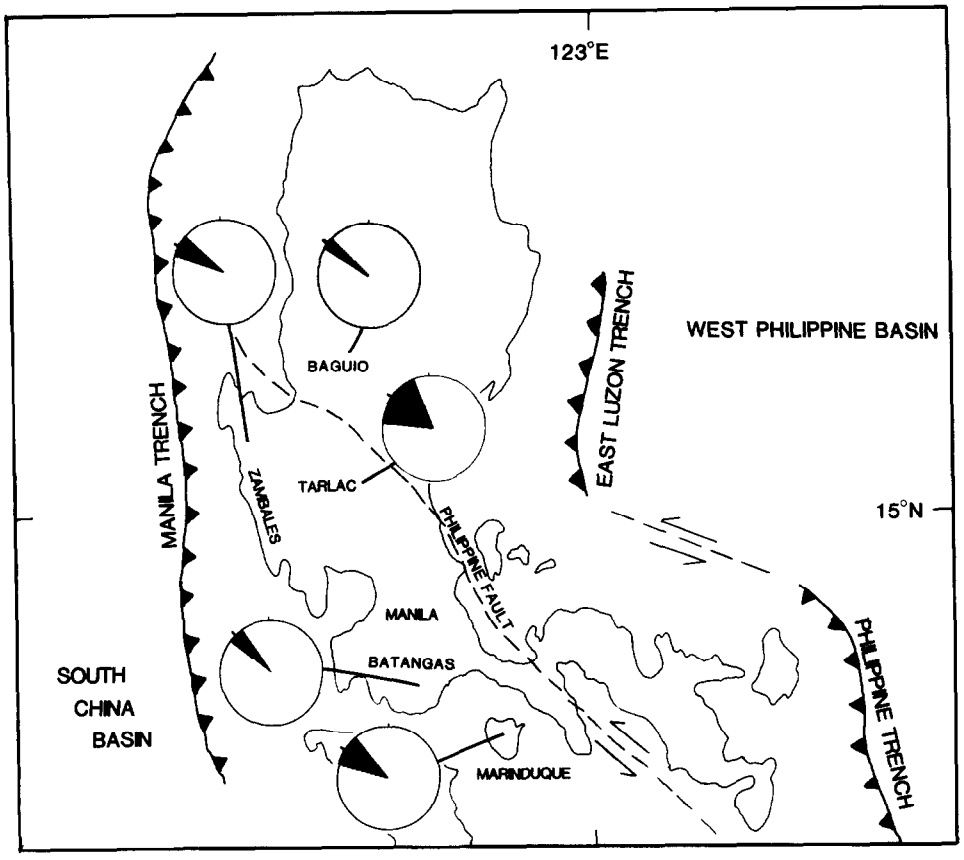


Fig. 19. Paleomagnetic results for Early and Mid-Miocene summarized on map of northern Philippines.

Given the interpretation of counterclockwise rotation in much of Luzon the results from Marinduque place that island as a part of a broad region of Miocene counterclockwise rotation and the general similarity of the results from Marinduque with those from Luzon is therefore to be expected. In contrast, on the basis of McCabe's most recent interpretation the results from Marinduque the observed declination anomalies would have been caused by opposite senses of rotation in the north and in Marinduque. The difficulties in such an interpretation are self-evident (Fig. 19).

Discussion and tectonic implications

The paleomagnetic results from Luzon suggest similar rotations of a substantial part of the island. From our limited data set, it is not possible to tell whether we are seeing rotations of small blocks within shear zones, or coherent rotation. The paleomagnetic data imply a small clockwise rotation of Luzon since Late Miocene, or Early Pliocene time. Immediately, prior to that, during the Middle Miocene, rapid counterclockwise rotation took place. Between Early Miocene and Early Oligocene times less rapid counterclockwise rotation accompanied by northerly motion took place. In the Late Eocene, there was additional counterclockwise rotation. Taken literally, the results suggest accompanying southerly motion. However, this is at the limit of resolution and so must await additional testing before it can be considered established. The tectonic implications of these results are now discussed.

The young clockwise rotation seen in Luzon is not detected in the central Philippines (McCabe et al., 1987). It is our interpretation that for a period of time in the Late Miocene and Pliocene, while the central Philippines were locked in their present orientation between two subduction zones, the northern Philippines were more strongly coupled to the Philippine Sea Plate and rotated with it. As

the East Luzon trench has progressed northward this coupling to the Philippine Sea Plate is lost. There must therefore have been a plate boundary in the form of a northward migrating transform, or region of diffuse deformation, between the part of Luzon rotating with the Philippine Sea Plate and that locked between the two subduction zones (Fuller, 1985). It is the clockwise rotation and westward component of relative motion of the northern Philippines with respect to the central Philippines which gives the Philippines their oroclinal like bend. A similar interpretation was suggested many years ago by Hamilton (1979).

The pre-Late Miocene results are consistent with the broad outline of the model presented by Holloway (1982). As parts of a proto-Philippine archipelago moved northwards and rotated coun-

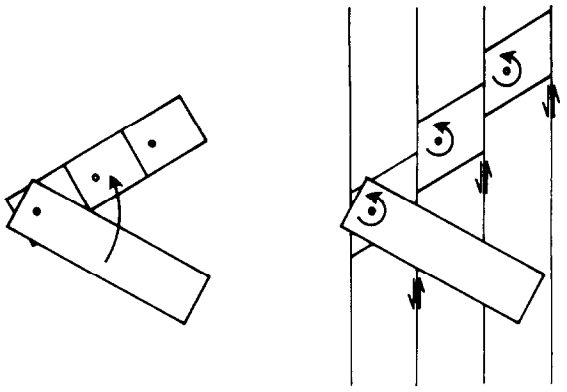


Fig. 20. End member models of paleomagnetic rotations. a. Rigid rotating beam. b. Sheared beam, with rotation of individual blocks, requiring internal deformation and adjustments at boundary, which are not shown.

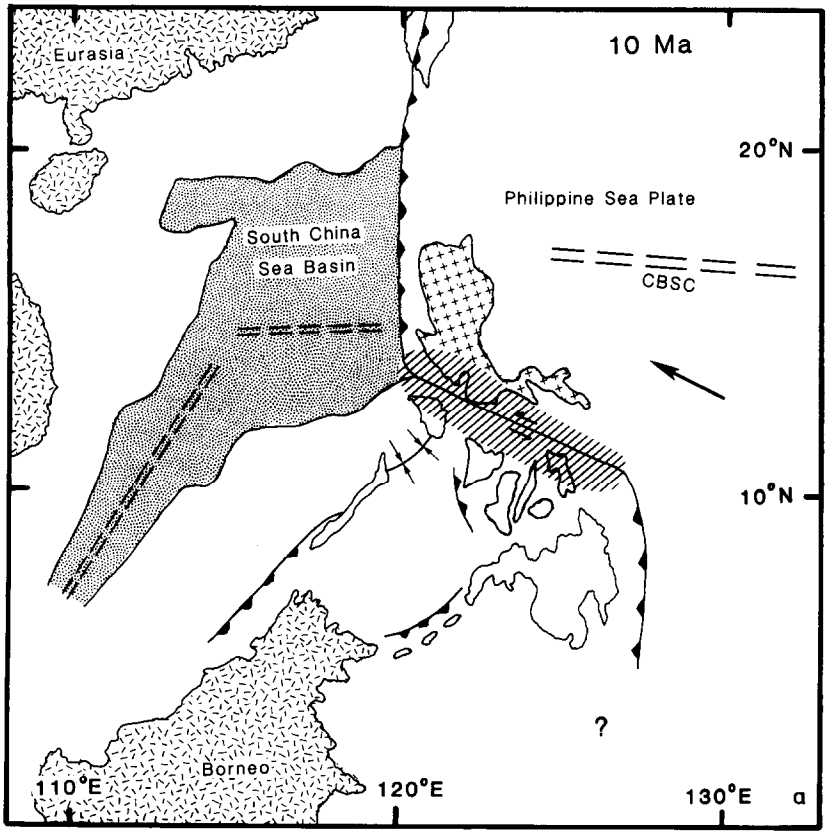
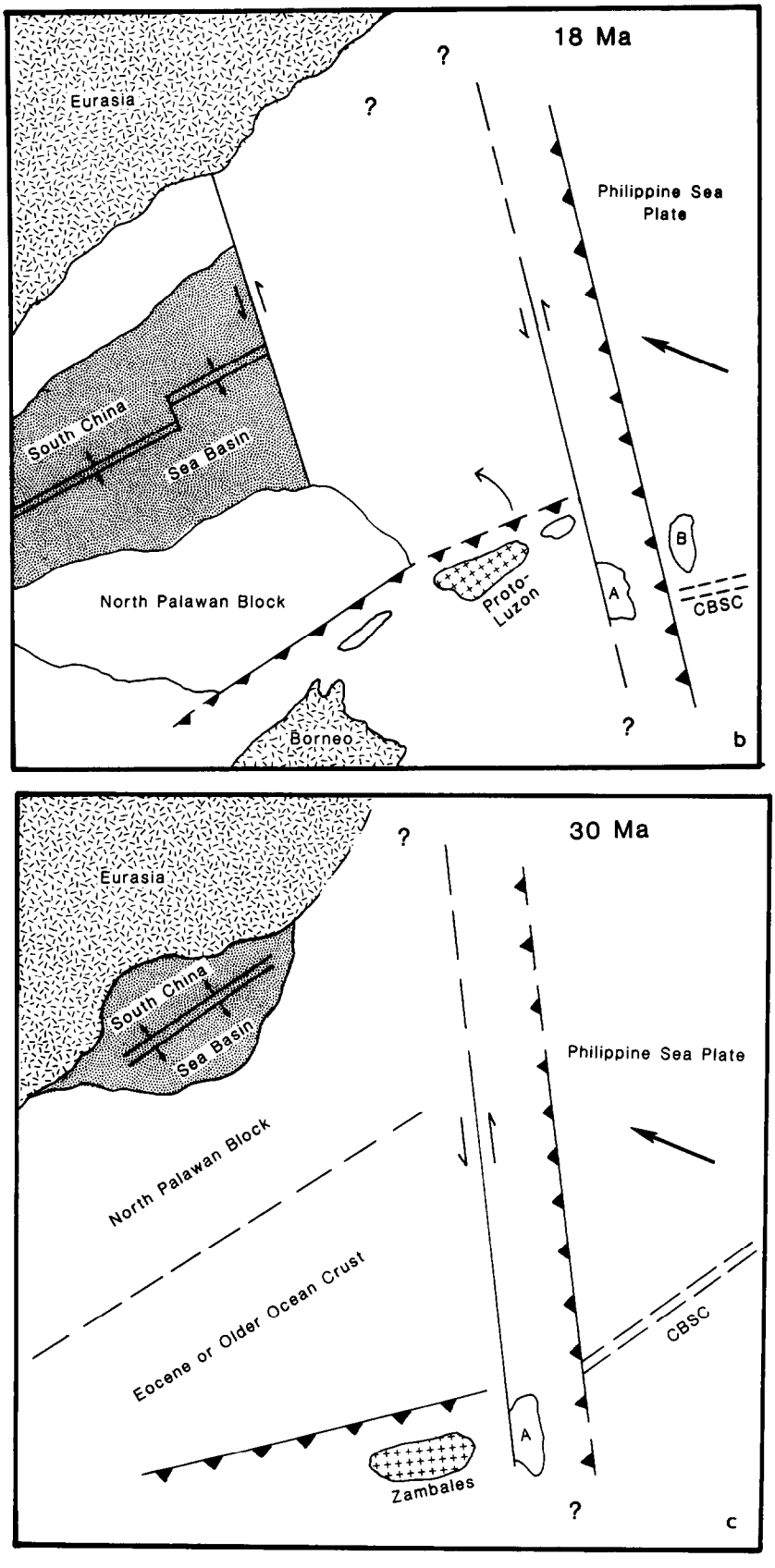


Fig. 21. Preliminary tectonic reconstructions consistent with paleomagnetic data. a. *10 Ma B.P.* Luzon is rotated counterclockwise to take account of the Pliocene–Late Miocene clockwise rotation. A region of diffuse deformation and sinistral shear is postulated between Luzon and the central Philippines. b. *18 Ma B.P.* Luzon is about to begin a period of rapid rotation around the North Palawan microcontinent. There may be fragments (*A*) moving northwards within the sinistral shear. There may be other topographic highs (*B*) similar to Benham Rise which remain on the Philippine Sea Plate. c. *30 Ma B.P.* Sinistral shear is established by Philippine Sea Plate motion. Land masses (*A*) may be moving northward in the shear. Zambales is on the leading edge of a northward moving plate which is over-riding ocean crust which originally lay to the south of China.


18 Ma
?
?
Eurasia
Philippine Sea Plate
South China Sea Basin
North Palawan Block
Proto-Luzon
A
B
CBSC
Borneo
?
b
30 Ma
?
Eurasia
South China Sea Basin
Philippine Sea Plate
North Palawan Block
Eocene or Older Ocean Crust
CBSC
A
Zambales
?
c

terclockwise, they collided with the Palawan microcontinent. The northern Philippines then rotated counterclockwise around this buttress throughout the Middle Miocene. To explain the motion of the proto-Philippine arc proposed by Holloway (1982), it is simplest to assume that the arc was on the leading edge of a plate itself moving northwards and rotating counterclockwise. This implies a pole of rotation somewhere relatively close and to the west. Whether this was a substantial plate, or simply an arc bounded by double subduction zones, such as parts of the present Philippine archipelago is not initially clear. Neither is it clear whether the plate rotated coherently, or was broken into a number of fragments. Indeed from paleomagnetic studies alone, this is indeterminate (Fig. 20).

A key aspect of Southeast Asia tectonics which is not discussed by Holloway (1982) is the role of the Philippine Sea Plate. At present the Philippine Sea Plate is considered to be rotating clockwise with respect to stable Asia about a pole of relative motion in the vicinity of Japan. A recent paleomagnetic study from Palau (Haston et al., 1988) suggests that the rotation has persisted at least since the Oligocene. The pole of rotation is not well determined, but is likely to be to the east of its present location in the geomagnetic reference frame. The Philippine Sea Plate motion defines a region of convergence and sinistral shear along its boundary with Asia.

The tectonics in such a region convergence is likely to be complex and their detailed history may be forever obscured by later events. Nevertheless, certain broad first order features may be predicted. Subduction will almost certainly be oblique. Figure 21 offers possible reconstructions for approximately 20 and 30 Ma. They are based upon Karig's earlier model, but recognize the need to place Zambales on a counterclockwise rotating plate, or microplate, to the west of the Philippine Sea Plate.

The sense of subduction on the northern boundary of the plate on which Zambales lies is consistent with the emplacement of the ophiolite (e.g., Schweller et al., 1983). The sense of subduction on the western boundary of the Philippine Sea Plate is less readily justified. At present, the Philippine trench subduction consumes Philippine Sea Plate. Indeed considerable amounts of Philippine Sea Plate must have been consumed, if the rotation proposed by Haston et al. (1988) has taken place. The mode of operation of this convergent system may have been analogous to that proposed by Hall (1987) to account for the evolution of Halmahera, which lies to the south of the Philippines. In this view, topographic highs on the leading edge of the Philippine Sea Plate, collide with the overiding plate and cause temporary changes in subduction, by trench jumps and polarity switches.

In addition, to the complication of changes in subduction regime, it seems very likely that the sinistral shear played an important role. It is here that the ideas of Fitch (1972), Beck (1980), Karig (1983) and Sarewitz and Karig (1986) are relevant. For example, Sarewitz and Karig (1986) suggest that the Philippine archipelago has been deformed and displaced in the left-lateral shear between the Philippine Sea Plate and Asia.

The model of deformation in such a region is likely to be described by some combination of two models discussed recently by McKenzie and Jackson (1983, 1986). The first derives from ideas of Freund (1974) and is a model of discontinuous deformation with a shear zone. The deformation is accommodated by rotation about a vertical axis of blocks which are rigidly attached to the two plates giving rise to the shear. The second is a continuously distributed deformation model and derives from hydrodynamic principles. It treats the blocks as floating in a medium within which there is a velocity gradient. In this model the blocks are not in contact with the boundaries. The rotation is then governed by the vorticity which can be calculated from the instanteous velocity vectors. The magnitude of the rotations which the results described in this paper reveal suggest that the second model is more appropriate with possible interactions between blocks carried by the fluid flow field and irregularities in the Asian Plate. Given this view, the Palawan microcontinent is seen as an obstacle around which parts of northern Luzon rotated rapidly.

We have presented a working hypothesis which attempts to explain the paleomagnetic data in

terms of a marriage of the ideas of Holloway (1982) and Sarewitz and Karig (1986). Local rotations on a smaller scale, may be involved in the process. Such rotations have by now been demonstrated in a number of localities (e.g., Luyendyk et al., 1980; Ron et al., 1984, and Wells and Coe, 1985). *Any* combination of rigid plate rotation and localized rotation which sums to give the correct declination anomaly is permitted by the paleomagnetic data. The key distinction appears to lie in the long term orientation of the shear zones. We suggest that parts of the Philippines have been caught in a broad and persistent shear zone since at least the Oligocene. However, we cannot yet determine the scale of the blocks which have behaved rigidly in this process.

We suggest that Zambales may have had a common early history with regions to the southwest. Parts of Sabah, Palawan, Mindoro and Zambales were on the leading edge of a plate, which rotated counterclockwise, as it moved northward during the early Tertiary. The actual trajectory is intermediate between that of the beam model and that of the shear zone. It was eventually disrupted by collision with the Palawan microcontinent in the Miocene, which appears to have given rise to clockwise rotations in Palawan and northern Sabah, while Zambales rotated counterclockwise around the same Palawan microcontinental block. To test this model we will make detailed collections moving to the southwest to establish the degree of coherence of the paleomagnetic results and implied motion.

Whether parts of the Philippines are derived by accretion from the Philippine Sea Plate is not clear from this study. We think that models which derive the whole Philippine archipelago from the leading edge of the Philippine Sea Plate (McCabe et al., 1987) are not promising because they are in conflict with the paleomagnetic data. However, that the Philippines are a part of a giant shear zone between the clockwise rotating Philippine Sea Plate and Asia, as suggested by Karig (1983) and by Sarewitz and Karig (1986) seems inescapable. Thus it is very likely that parts of the Philippines have been accreted from the Philippine Sea Plate in the manner proposed further for parts of Halmahera by Hall (1987). Haston et al. (1988) have demonstrated that the Philippine Sea Plate has rotated clockwise since at least the Mid-Oligocene. In such an environment of oblique subduction, the role of strike-slip motion in the upper plate along proto-Philippine faults is likely to be important, as pointed out by Karig (1983). We plan to test these ideas by paleomagnetic studies in the southeastern parts of the Philippines, to see if there are accreted fragments with a paleomagnetic record of Philippine Sea Plate history.

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